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Arpan Adhikari

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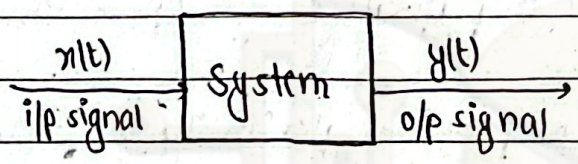
Signal

A signal is defined as a function of one or more variables that carries info. A signal is a physical quantity that varies with time or in general any other independent variable.

system

A system is defined as a physical device that perform an operation on a signal.

It is an entity that manipulates one or more i/p signals to perform a function which results in a new o/p signal.



Signal Processing

Passing the signal to the system is known as processing. Most of the signals encountered in science & technology are analog in nature.

- If the system is processed directly in its analog form, then the processing is analog signal processing.

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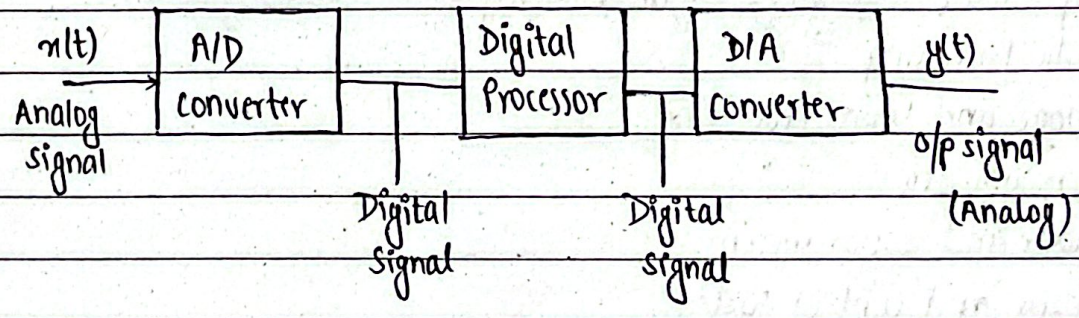
Digital signal Processing



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- Processing of digital signal on a digital computer is known as DSP.
- The block diagram for digital signal processing is shown in above figure that consists of Analog to Digital converter, digital processor and Digital to Analog converter.
- To perform the processing digitally, there is need for an interface between the analog signal and the digital processor. This interface is known as analog to digital converter.
- The o/p of analog to digital converter is the digital signal that is appropriate input to the digital processor.
- The digital signal processor maybe a digital computer, or small microprocessor programmed to perform design operation on the i/p signal.
- Programmable machine provides flexibility to change the signal processing operations through a change in the software. On the other hand, when signal processing operations are well defined, a hardware implementation are more convenient and reliable.
- The digital to analog converter is used to provide analog signal from the digital o/p of the digital signal processor.

Application of Digital signal Processing

1. Audio Processing
2. Image and Video Processing
3. Communications
4. Radar and sonar systems
5. Sensor and control system
6. Medical signal Processing
7. Financial and scientific analysis

Advantages of Digital signal Processing

1. High accuracy and precision in processing compared to analog methods.
2. Flexibility : modifications can be made easily by changing software instead of hardware.
3. Error detection and correction features are readily available with digital signals.
4. Digital signals are less susceptible to noise and interference.
5. Ease of storage, transmission and encryption of digital data.
6. Universal compatibility and easy interfacing with other digital devices.

Disadvantages of Digital signal Processing

1. Requires additional hardware (ADC and DAC) to convert between analog and digital signals, increasing overall system complexity.
2. Higher power consumption compared to analog processing due to intensive computations and internal resources.
3. Can be more expensive due to costly digital signal processors.
4. May introduce quantization and round-off errors during analog-to-digital conversion, potentially losing fine details.
5. System are generally more complex, often requiring skilled personnel for design and maintenance.

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1) Sampling

$t = nT$

$T = \frac{1}{F_s}$

$t = \frac{n}{F_s}$

Nyquist Rate $= 2f_{max}$

2) Quantization

Δ (Resolution)

or

$\text{Stepsize} = \frac{x_{max} - x_{min}}{L-1}$

where, L = No. of Levels

x_{max} = Maximum Amplitude

$x_{min} = -x_{max}$

3) Encoding

$b \geq \log_2(L)$

where,

b = no. of bits per sample

L = No. of Levels

$\text{sampling freq.} = \frac{\text{Bit rate}}{\text{No. of bits/sample}}$

$\text{Folding frequency} = \frac{F_s}{2}$

(aliasing अर्थात Recover नहीं होगा)

For signals not having aliasing,

$-1/2 < f < 1/2$

↑ No aliasing → recover!

Q) Consider a signal $x_a(t) = 10\cos 2\pi 1000t + 5\cos 2\pi 5000t$ is to be sampled.

(i) Determine the Nyquist rate for this signal.

(ii) If the signal is sampled at 4KHz, will the signal be recovered from this sample?

Soln

$x_a(t) = 10\cos 2\pi 1000t + 5\cos 2\pi 5000t$

Comparing with standard eqn. $x_a(t) = A_1\cos 2\pi f_1 t + A_2\cos 2\pi f_2 t$,

we get,

$A_1 = 10, f_1 = 1000, A_2 = 5, f_2 = 5000$

So, Max^m Frequency (f_{max}) = 5000.

(i) Nyquist rate $= 2 \times f_{max}$

$= 2 \times 5000 = 10000 = 10\text{KHz}$ Ans

(ii) $F_s = 4\text{KHz}$

Signal can be written as,

$x(n) = 10\cos 2\pi 1000 \times \frac{n}{F_s} + 5\cos 2\pi 5000 \times \frac{n}{F_s}$

$= 10\cos 2\pi \frac{1000 \times n}{4000} + 5\cos 2\pi \frac{5000 \times n}{4000}$

$= 10\cos 2\pi\left(\frac{n}{4}\right) + 5\cos 2\pi\left(\frac{5}{4}\right)n$

$= 10\cos 2\pi\left(\frac{1}{4}\right)n + 5\cos 2\pi\left(1+\frac{1}{4}\right)n$

$= 10\cos 2\pi\left(\frac{1}{4}\right)n + 5\cos 2\pi\left(\frac{1}{4}\right)n$

$= 15\cos 2\pi\left(\frac{1}{4}\right)n$

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As $f = \frac{1}{4}$, that lies betⁿ $-1/2$ to $1/2$. Hence, the signal can be recovered from sample. Ans

Q) A digital communication link carries binary coded word representing sample of an input signal

$x_a(t) = 3\cos 600\pi t + 2\cos 800\pi t$

The link is operated at 10,000 bits per second and each input sample is quantized into 1024 different voltage levels.

(i) What is the sampling frequency & folding frequency?
(ii) What is the Nyquist rate?
(iii) What are the frequencies in resulting discrete time signal.
(iv) What is resolution?

Soln

No. of levels (L) = 1024

No. of bits per sample (b) = $\log_2(L)$
$= \log_2(1024)$
$= 10$ bits/sample

(i) Sampling freq. = $\frac{\text{Bit rate}}{\text{No. of bits/sample}}$

$= \frac{10{,}000}{10}$

$\therefore F_s = 1000$ Hz

Folding freq. (F.F) = $\frac{F_s}{2} = \frac{1000}{2}$

$= 500$ Hz Ans

For composite system,
X_{max} = sum of amplitude of all signals

(ii) $x_a(t) = 3\cos 600\pi t + 2\cos 800\pi t$

comparing with $x_a(t) = A_1\cos 2\pi f_1 t + A_2\cos 2\pi f_2 t$, ($\omega = 2\pi f$)

$A_1 = 3$, $A_2 = 2$, $f_1 = 300$, $f_2 = 400$

~~Nyquist rate~~

So, $f_{max} = 400$

And, Nyquist rate = $2 \times f_{max} = 2 \times 400$
$= 800$ Hz Ans

(iii) In discrete time signal,

$x(n) = 3\cos 600\pi \frac{n}{F_s} + 2\cos 800\pi \frac{n}{F_s}$

$= 3\cos \frac{600\pi}{1000} n + 2\cos \frac{800\pi}{1000} n$

$= 3\cos 2\pi\left(\frac{3}{10}\right)n + 2\cos 2\pi\left(\frac{4}{10}\right)n$

(comparing with standard discrete time signal eqⁿ $x(n) = A_1\cos 2\pi f_1 n + A_2\cos 2\pi f_2 n$,

$A_1 = 3$, $A_2 = 2$, $f_1 = 0.3$ Hz, $f_2 = 0.4$ Hz Ans

(iv) $L = 1024$

x_{max} (Maxᵐ Amplitude) $= 3 + 2 = 5$

$\therefore x_{min} = -x_{max} = -5$

Now,

Resolution/step size = $\frac{x_{max} - x_{min}}{L - 1}$

$= \frac{5 - (-5)}{1024 - 1}$

$= \frac{10}{1023}$

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Elementary signals are basic standard signals used as fundamental components to construct and analyze complex signals in signal processing.

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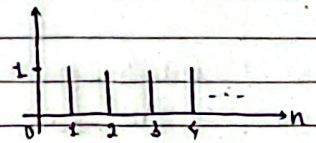
Elementary Discrete time signals (sequence)

1. Unit step sequence

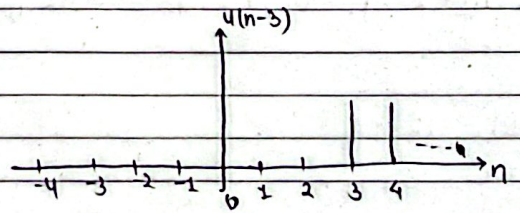


$u(n) = 1$ for $n \geq 0$
0 for $n < 0$

Shifted version is $u(n-k)$.

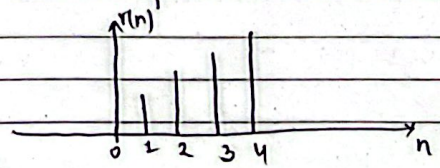
- $u(n-k)$ → Right shift by k units
- $u(n+k)$ → Left shift by k units
- $u(-n-k)$ → Left shift by k units
- $u(-n+k)$ → Right shift by k units

(Advance → day)



2. Unit Ramp Sequence

$r(n) = n$ for $n \geq 0$
0 for $n < 0$



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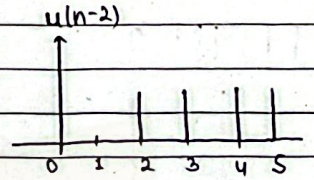
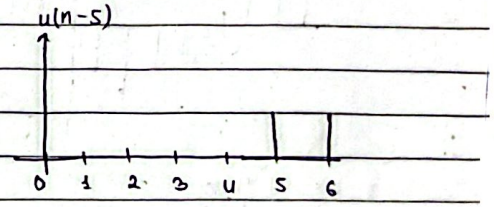
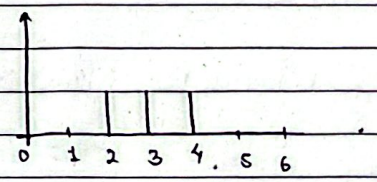
(a) $\{r(n-k) = n$ for $n \geq k$
0 for $n < 0\}$

Q) Plot $x(n) = u(n-2) - u(n-5)$



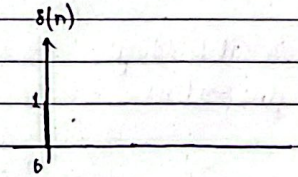


$u(n-2) - u(n-5)$:



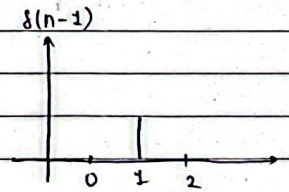
3. Unit Impulse Sequence

$\delta(n) = 1$ for $n = 0$
0 for $n \neq 0$



Shifted version is $\delta(n-k)$

$\delta(n-k) = 1$ for $n = k$
0 for $n \neq k$



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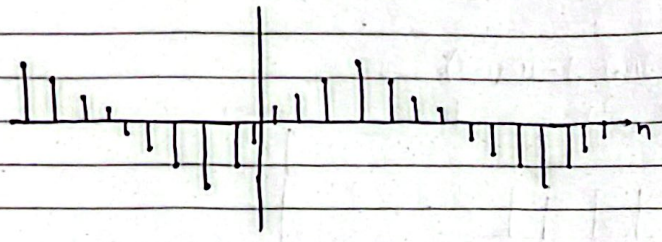
4. Sinusoidal sequence

$x(n) = A \sin \omega n \qquad -\infty < n < \infty$



5. Time shifting

$y(n) = x(n-K)$

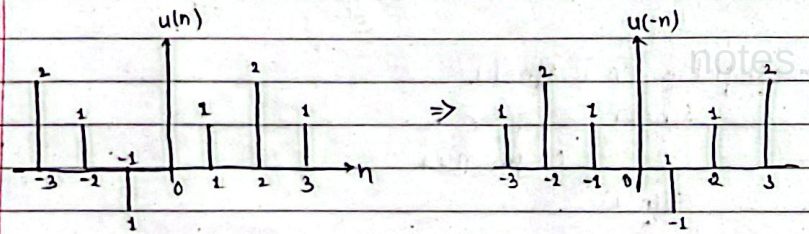
Basic Operation on Discrete Time signals (sequence)

1. Time shifting

$y(n) = x(n-k)$

2. Time Reversal (Folding)

$y(n) = x(-n)$



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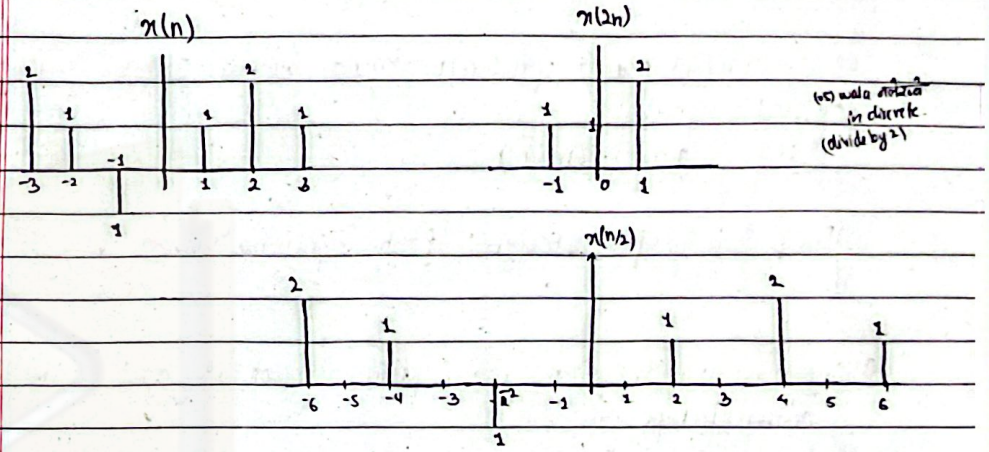
3. Time scaling

$y(n) = x(an)$

{ a>1, signal compress
a<1, signal expand } amp. constant



Classification of Discrete time signal

1. Deterministic & Random signal

A signal exhibiting no uncertainty of its magnitude and phase at any given instant of time is called deterministic signal. A deterministic signal can be expressed by mathematical eqn at any time and its nature and amplitude at any time can be predicted.

E.g. Sinusoidal signal, Exponential signal, Unit step signal

A signal characterized by uncertainty about its occurrence is called random signal. A random signal cannot be represented by any mathematical eqn

Eg. Noise

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2 Periodic & Aperiodic Signal

A signal which has a definite pattern and repeats itself at regular interval of time is called a periodic signal.

For periodicity, $x(n) = x(n+N)$

where, N = Fundamental period

A signal which doesn't repeat after regular interval of time is called aperiodic signal.

Here, $x(n) \neq x(n+N)$

{For periodicity of Discrete signal, $f = \frac{k}{N}$ = Rational Number }

Q) Check whether the given signal is periodic or not. If yes, then find fundamental period.

a) $x(n) = \sin(0.02\pi n)$

Soln Comparing with standard eqⁿ $x(n) = \sin \omega n$,

$$\omega = 0.02\pi$$

$$\text{or, } 2\pi f = 0.02\pi$$

$$\therefore f = 0.01 = \frac{1}{100} \quad \left(\text{in form of } \frac{k}{N}\right)$$

Hence, signal is periodic and fundamental period = 100.

b) $x(n) = \sin\left(\frac{2\pi n}{3}\right) + \cos\left(\frac{2\pi n}{5}\right)$

Comparing with standard eqⁿ $x(n) = \sin\omega_1 n + \cos\omega_2 n$,

$$\omega_1 = \frac{2\pi}{3} \qquad \omega_2 = \frac{2\pi}{5}$$

$$\therefore f_1 = \frac{1}{3}\left(\frac{k_1}{N_1}\right) \qquad \therefore f_2 = \frac{1}{5}\left(\frac{k_2}{N_2}\right)$$

{Composite signal : check periodicity = $\frac{N_1}{N_2}$ => Rational No.
Fundamental period : LCM of individual period }

For composite signal,

$$\frac{N_1}{N_2} = \frac{3}{5} \quad \text{(Rational Number)}$$

So, it is periodic.

Fundamental period = $3 \times 5 = 15$ [(3,5) अरुको भए => 5] LCM.

c) $x(n) = \cos\left(\frac{\pi}{2} + 0.3n\right)$

Comparing with standard eqⁿ $x(n) = \cos(\omega n + \phi)$,

$$\omega = 0.3$$

$$\text{or, } f = \frac{0.3}{2\pi} \quad \text{(Not rational, so aperiodic)}$$ Ans

d) $x(n) = \cos\left(\frac{n}{6}\right) \cdot \cos\left(\frac{\pi n}{6}\right)$

$$= \frac{1}{2}\left[\cos\left(\frac{n}{6} + \frac{\pi n}{6}\right) + \cos\left(\frac{n}{6} - \frac{\pi n}{6}\right)\right]$$

Comparing with $x(n) = \cos(\omega_1 n) + \cos(\omega_2 n)$,

$$\omega_1 = \frac{1}{6} \qquad \omega_2 = \frac{\pi}{6}$$

$$\therefore f_1 = \frac{1}{2\pi 6} \text{ (irrational) (aperiodic)} \qquad f_2 = \frac{1}{12} \text{ (rational) (periodic)}$$

Product of aperiodic & periodic is aperiodic.

e) $e^{j\left(\frac{\pi n}{2}\right)}$

Comparing with $e^{j\omega n} = x(n)$

$$\omega = \frac{\pi}{2}$$

$$\text{or, } f = \frac{1}{4} \quad \text{(Rational). So, it is periodic}$$

Fundamental period = 4 Ans

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3. Energy & Power signal

A signal is said to be Energy signal if and only if its total energy E is finite and average power P is zero.

The energy of discrete time signal $x(n)$ is given by,

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

A signal is said to be Power signal if the average power is finite and the energy is infinite.

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

4. Causal & Non-causal signal

A discrete time signal $x(n)$ is said to be causal if $x(n) = 0$ for $n < 0$

A discrete time signal $x(n)$ is said to be non-causal if $x(n) = 0$ for $n > 0$.

5. Even & Odd signal

A discrete time signal $x(n)$ is said to be symmetric /Even signal) if it satisfies $x(n) = x(-n)$ for all n.

A discrete time signal $x(n)$ is said to be odd/ asymmetric /anti-symmetric if it satisfies $x(n) = -x(-n)$ for all n.

$$\left\{ \begin{array}{l} x_e(n) = \frac{1}{2}[x(n) + x(-n)] \\ x_o(n) = \frac{1}{2}[x(n) - x(-n)] \end{array} \right\}$$

#same inam wala signal ma (+,-,x)...operation garne filter,

Q) Determine even & odd signal of given signal : $x(n) = \{-4, 2, \underset{\uparrow}{3}, 1, 2\}$

$$x(-n) = \{2, 1, \underset{\uparrow}{3}, 2, -4\}$$

Even signal $x_e(n) = \frac{1}{2}[x(n) + x(-n)]$

$= \frac{1}{2}\{-2, 3, 6, 3, -2\}$

$= \{-1, 1.5, 3, 1.5, -1\}$ Ans

Odd signal $x_o(n) = \frac{1}{2}[x(n) - x(-n)]$

$= \frac{1}{2}\{-6, 1, 0, -1, 6\}$

$= \{-3, 0.5, 0, -0.5, 3\}$ Ans

Classification of Discrete Time System

1. Static and Dynamic system

A system is said to be static or memoryless if the response is due to the present input alone i.e. for static system, the o/p at any instant n depends only on the input applied at that instant n, but not on the past or future values of input or past values of output.

A system is said to be dynamic or memory system if the response depends on past or future i/p or past o/p.

$y(n) = x(n-1) + y(n-4)$ Past i/p → dynamic

~~$y(n) = x(n-1) + x(n+2)$~~

$y(n) = x(n) + x(n-2)$

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2. Causal & Non-causal system

A system is said to be causal if the o/p of the system at any instant n depends only on the present & past values of the i/p but not on the future values.

The impulse response of a causal system is given by,

$$h(n) = 0 \text{ for } n < 0$$

A system is said to be non-causal if the o/p of the system at any instant n depends on future i/p.

E.g. $y(n) = x(2n)$

3. Linear & Non-linear system

A system is said to be linear which obeys the principle of superposition and principle of homogenity.

A system is said to be non-linear which doesn't obey the principle of superposition & principle of homogenity.

Q) Check whether the following system is linear or non-linear. $y(n) = n^2 x(n)$

Soln

Let $x_1(n)$ be i/p and $y_1(n)$ be the o/p.

then, $y_1(n) = n^2 x_1(n)$ —(i)

similarly,

$$y_2(n) = n^2 x_2(n) \quad \text{—(ii)}$$

Now,

$$ay_1(n) + by_2(n) = an^2 x_1(n) + bn^2 x_2(n)$$
$$= n^2 [a x_1(n) + b x_2(n)] \quad \text{—(iii)}$$

$$y_3(n) = T[ax_1(n) + bx_2(n)]$$
$$y_3(n) = n^2 [ax_1(n) + bx_2(n)] \quad \text{—(iv)}$$

Eqn (iii) & (iv) is equal. Hence, the system is linear. Ans

Q) Check linearity : $y(n) = x(n) + \frac{1}{2x(n-2)}$

Soln

Let $x_1(n)$ be i/p & $y_1(n)$ be o/p.

then, $y_1(n) = x_1(n) + \frac{1}{2x_1(n-2)}$

Similarly, $y_2(n) = x_2(n) + \frac{1}{2x_2(n-2)}$

Now, $ay_1(n) + by_2(n) = a\left[x_1(n) + \frac{1}{2x_1(n-2)}\right] + b\left[x_2(n) + \frac{1}{2x_2(n-2)}\right]$ —(i)

Now, $y_3(n) = T[ax_1(n) + bx_2(n)]$

$$= T\ ax_1(n) + a\frac{1}{2x_1(n-1)}$$

$$= ax_1(n) + bx_2(n) + \frac{1}{2[ax_1(n-2) + bx_2(n-2)]} \quad \text{—(ii)}$$

Eqn (i) & (ii) is not equal. So, system is non-linear. Ans

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4. Time varient & Time invarient System:

A system is said to be time varient if its i/p o/p characteristics changes with time.

A system is said to be time invarient if its i/p o/p characteristics donot change with time.

Steps to determine whether the given system is time varient/invarient:

(i) Let $x(n)$ be the input. The o/p is obtained by shifting the i/p by K unit. The o/p is represented by $y(n,K)$.

(ii) The o/p $y(n)$ is shifted by K units and n is also shifted by K units throughout the eqn. The o/p is represented by $y(n-K)$.

(iii) If $y(n,K) = y(n-K)$, then the system is time invarient.

Q) Check time varience: $y(n) = x(n/2)$

Soln

Shifting the i/p by K units,

$$y(n,K) = x\left(\frac{n}{2} - K\right)$$

Shifting Now, $y(n-K) = x\left(\frac{n-K}{2}\right)$

Here, $y(n,K) \neq y(n-K)$, so system is time varient. Ans

Q) Check time varience: $y(n) = x(n) + x(n-2)$.

Soln

Here, $y(n,K) = x(n-K) + x(n-K-2)$

Now,

$$y(n-K) = x(n-K) + x(n-K-2)$$

So, $y(n,K) = y(n-K)$. So, system is time invarient Ans

Q) $x(n) + nx(n-2)$

Soln

$$y(n,K) = x(n-K) + nx(n-K-2)$$

Again,

$$y(n-K) = x(n-K) + (n-K)x(n-K-2)$$

So, $y(n,K) \neq y(n-K)$. Time varient.

Q) Check whether the given system

(i) static/Dynamic (ii) causal/Non-causal (iii) linear/Non-linear (iv) time varient/invarient

$y(n) = a^n x(n)$

Soln

(i) The o/p depends on present i/p only so, static system.

(ii) Depends on present i/p only. so, causal system.

(iii) Let $y_1(n)$ be o/p for i/p $x_1(n)$.

$$y_1(n) = a^n x_1(n)$$

similarly, $y_2(n) = a^n x_2(n)$

$$by_1(n) + cy_2(n) = ba^n x_1(n) + ca^n x_2(n)$$
$$= a^n [bx_1(n) + cx_2(n)] \quad \text{—— (i)}$$

$$y_3(n) = T[bx_1(n) + cx_2(n)]$$
$$= a^n [bx_1(n) + cx_2(n)] \quad \text{—— (ii)}$$

As eqn (i) & (ii) are equal. system is linear.

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(iv) $y(n,k) = a^n x(n-k)$

$y(n-k) = a^{n-k} x(n-k)$

Here, $y(n,k) \neq y(n-k)$. So, time varient.

5. Stable & Unstable System (BIBO system)

A system is said to be stable if and only-if every bounded i/p produces bounded o/p.

If the system doesn't produce bounded o/p for bounded i/p, the system is unstable.

Imp

Theorem:

For The necessary and sufficient condition for a discrete time system to be stable is given by $\left|\sum_{n=-\infty}^{\infty} h(n)\right| < \infty$ for $n<0$

Consider a LTI system with $x(n)$ as input and $y(n)$ as output. The input and output of the system are related by convolution integral given by

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k) \quad \text{——(i)}$$

Taking the absolute value on both sides of eqn (i), we get,

$$|y(n)| = \left|\sum_{k=-\infty}^{\infty} x(k)\, h(n-k)\right| \quad \text{——(ii)}$$

Using the fact that the absolute value of sum of product of two terms is always less than or equal to sum of product of their absolute value.

$$\left|\sum_{k=-\infty}^{\infty} x(k)\, h(n-k)\right| \leq \sum_{k=-\infty}^{\infty} |x(k)|\,|h(n-k)| \quad \text{——(iii)}$$

If the i/p $x(k)$ is bounded i.e. there exists finite no. m such that

$$|x(k)| \leq m < \infty \quad \text{——(iv)}$$

Substituting value of eqn (iv) to eqn (ii);

$$|y(n)| \leq m \left|\sum_{k=-\infty}^{\infty} h(n-k)\right|$$

Changing the variable $(n-k)$ by m, the o/p is bounded only if

$$\left|\sum_{m=-\infty}^{\infty} h(m)\right| < \infty$$

Replacing m by n, we get,

$$\left|\sum_{n=-\infty}^{\infty} h(n)\right| < \infty$$

Eg 3

Causality of System

Theorem: A LTI system is causal if

$$h(n) = 0 \text{ for } n<0$$

The convolution of two signals $x(n)$ & $h(n)$ is given by,

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k) \quad \text{——(i)}$$

The above eqn (i) can also be written as,

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k) \quad \text{——(ii)}$$

The above eqn (ii) can be splitted into two parts,

$$y(n) = \sum_{k=-\infty}^{-1} h(k)\, x(n-k) + \sum_{k=0}^{\infty} h(k)\, x(n-k) \quad \text{——(iii)}$$

Now expanding eqn (iii), we get,

$$y(n) = \ldots + h(-3)x(n+3) + h(-2)x(n+2) + h(-1)x(n+1) + h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \ldots$$

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Let us consider that the present condition is at n=0, then

$x(n)$ = present i/p

$\cdots x(n+3), x(n+2), x(n+1)$ = Future i/p

$\cdots x(n-1), x(n-2), x(n-3)$ = Past i/p

For the system to be causal, the o/p of the system depends on the present and past i/p, not on the future i/p.

Therefore, the term $h(-1), h(-2), h(-3) \ldots$ should be equal to 0.

So, $h(n) = 0$ for $n < 0$ Proved

Linear Convolution or Convolution Sum

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Graphical Method:

Steps:

For all methods

1. The starting inden of $y(n)$ is equal to sum of starting inden of $x(n)$ and $h(n)$
2. The ending inden of $y(n)$ equals to sum of ending inden of $x(n)$ and $h(n)$.
3. The length L_y is related to L_x and L_h by
   $L_y = L_x + L_h - 1$
4. Express both the sequences $x(n)$ and $h(n)$ in terms of inden k.
5. Fold $h(k)$ about $k=0$ to obtain $h-k$ and shift to the right if n is positive and to the left if n is negative.
6. Multiply the two sequences $x(k)$ and $h(n-k)$ element by element and sum the product to get $y(n)$.

Q) Find the linear convolution of signal given below

$x(n) = \{\overset{0}{4}, \overset{1}{2}, \overset{2}{1}, \overset{3}{3}\}$ (Arrow denotes 1st element 0 inden)

$h(n) = \{\overset{-1}{1}, \overset{0}{2}, \overset{1}{2}, \overset{2}{1}\}$

↑

Starting inden of $y(n) = 0 - 1 = -1$

Ending inden of $y(n) = 3 + 2 = 5$

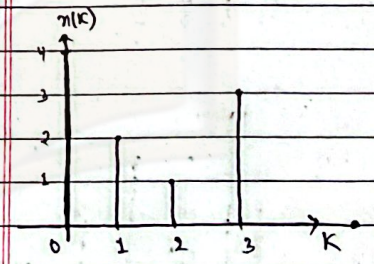
Length $L_y = L_x + L_h - 1$
$= 4 + 4 - 1$
$= 7$

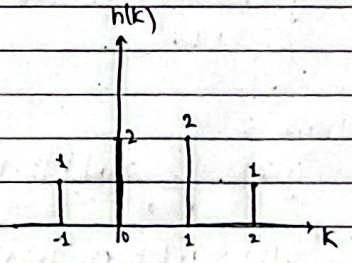
{ L_x => Length of x => No. of elements
L_y => " " y => No. of elements }

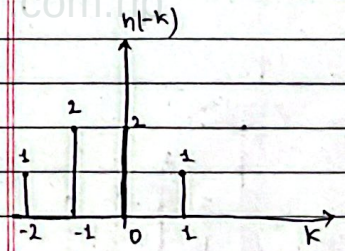
$x(k) = \{4, 2, 1, 3\}$

$h(k) = \{\overset{-1}{1}, 2, 2, 1\}$

↑







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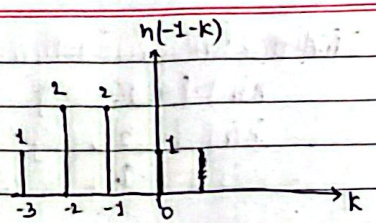
$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$$

For $n=-1$,

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)\,h(-1-k)$$

$$y(-1) = 4\times1 = 4$$



For $n=0$,

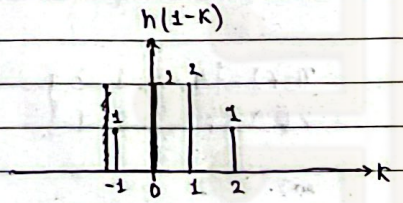
$$y(0) = \sum_{k=-\infty}^{\infty} x(k)\,h(-k)$$

$$y(0) = 4\times2 + 2\times1 = 10$$

For $n=1$,

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)\,h(1-k)$$

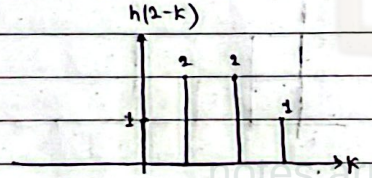
$$y(1) = 4\times2 + 2\times2 + 1\times1 = 13$$



For $n=2$,

$$y(2) = \sum_{k=-\infty}^{\infty} x(k)\,h(2-k)$$

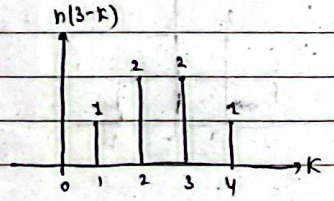
$$y(2) = 4\times1 + 2\times2 + 1\times2 + 3\times1 = 13$$



For $n=3$,

$$y(3) = \sum_{k=-\infty}^{\infty} x(k)\,h(3-k)$$

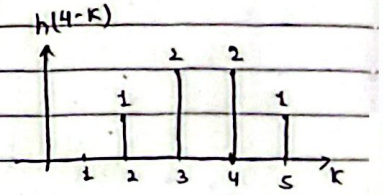
$$y(3) = 2\times1 + 2\times1 + 2\times3 = 10$$



For $n=4$,

$$y(4) = \sum_{k=-\infty}^{\infty} x(k)\,h(4-k)$$

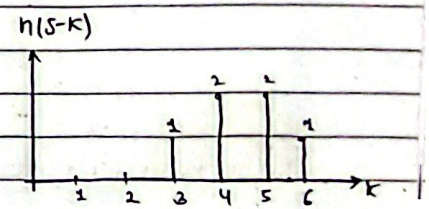
$$y(4) = 1\times1 + 3\times2 = 7$$



For $n=5$,

$$y(5) = \sum_{k=-\infty}^{\infty} x(k)\,h(5-k)$$

$$y(5) = 3\times1 = 3$$



Hence,

$$y(n) = \{4, 10, 13, 13, 10, 7, 3\}$$

Ans

Q) Find linear convolution of signal given alongside, $x(n) = \{3, 1, 1, 2\}$, $h(n) = \{2, 2, 1, 1\}$

Soln

Starting index of $y(n) = -1 + (-1) = -2$

Ending index of $y(n) = 2 + 2 = 4$

Now, $L_y = L_x + L_h - 1$

$= 4 + 4 - 1$

$= 7$

Here, $x(k) = \{3, 1, 1, 2\}$

$h(k) = \{2, 2, 1, 1\}$

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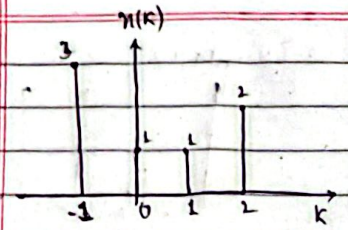
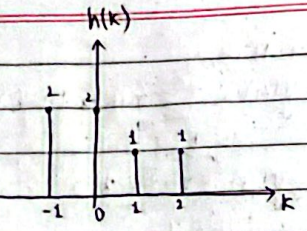
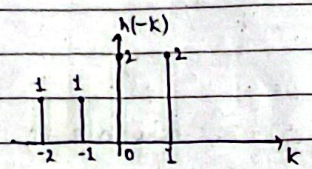






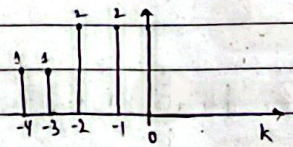
To find linear convolution, we have, $y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$

For n = -2,

$y(-2) = \sum_{k=-\infty}^{\infty} x(k)h(-2-k)$

$y(-2) = 3 \times 2$

$= 6$

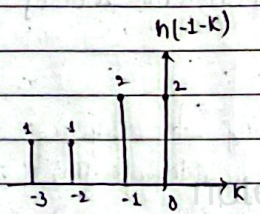


For n = -1,

$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$

$y(-1) = 3 \times 2 + 1 \times 2$

$= 8$



For n = 0,

$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$

$y(0) = 1 \times 3 + 1 \times 2 + 1 \times 2$

$= 7$

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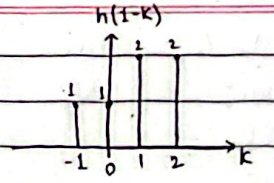
For n = 1,

$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$

$y(1) = 3 \times 1 + 1 \times 1 + 1 \times 2 + 2 \times 2$

$= 10$ //

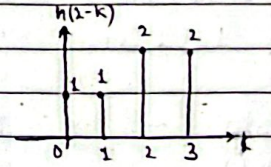


For n = 2,

$y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$

$y(2) = 1 \times 1 + 1 \times 1 + 2 \times 2$

$= 6$

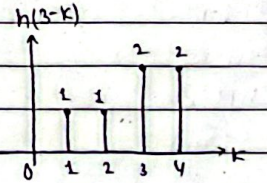


For n = 3,

$y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$

$y(3) = 1 \times 1 + 2 \times 1$

$= 3$

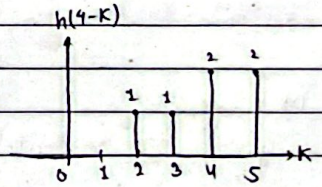


For n = 4,

$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$

$y(4) = 2 \times 1$

$= 2$



Hence,

$y(n) = \{6, 8, 7, 10, 6, 3, 2\}$

↑

Ans

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Tabular Array Method

$x(n) = \{4, 2, 1, 3\}$

$h(n) = \{1, 2, 2, 1\}$ (↑ at 1)

starting index = 0-1 = -1

Ending index = 3+2 = 5

$L_y = L_x + L_h - 1 = 4+4-1 = 7$

Index	-4	-3	-2	-1	0	1	2	3	4	5	6
x(k)					4	2	1	3			
h(k)				1	2	2	1				
h(-k)			1	2	2	1					
h(-1-k)		1	2	2	1						
h(-k)			1	2	2	1					
h(1-k)				1	2	2	1				
h(2-k)					1	2	2	1			
h(3-k)						1	2	2	1		
h(4-k)							1	2	2	1	
h(5-k)								1	2	2	1

At n = -1,

$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$

$\therefore y(-1) = 1 \times 4 = 4$

At n = 0,

$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$

$\therefore y(0) = 4\times2 + 2\times1 = 10$

At n = 1,

$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$

$\therefore y(1) = 4\times2 + 2\times2 + 1\times1 = 13$

At n = 2,

$y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$

$\therefore y(2) = 4\times1 + 2\times2 + 2\times1 + 3\times1 = 13$

At n = 3,

$y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$

$\therefore y(3) = 2\times1 + 2\times1 + 3\times2 = 10$

At n = 4,

$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$

$\therefore y(4) = 1\times1 + 2\times3 = 7$

At n = 5,

$y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k)$

$\therefore y(5) = 3\times1 = 3$

Hence, $y(n) = \{4, 10, 13, 13, 10, 7, 3\}$ Ans (↑ at 10)

Q) Find linear convolution of the signal,

$x(n) = \sum_{k=0}^{3} \delta(n-k)$

$h(n) = 2^n [u(n) - u(n-3)]$

Soln

$x(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3)$

So, $x(n) = \{1, 1, 1, 1\}$ (↑ at 1)

$h(n) = 2^n [u(n) - u(n-3)]$

$= 2^n \{1, 1, 1\}$ (↑ at 1)

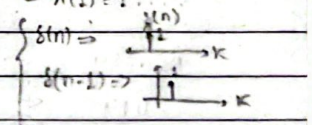
$= \{2^0\times1, 2^1\times1, 2^2\times1\}$

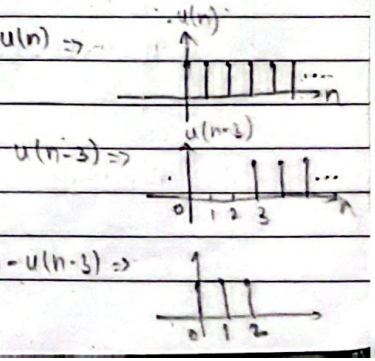
$h(n) = \{1, 2, 4\}$ (↑ at 1)

So, $x(k) = \{1, 1, 1, 1\}$ (↑ at 1)

$h(k) = \{1, 2, 4\}$ (↑ at 1)

At n=0,
$x(0) = \delta(0)+\delta(-1)+\delta(-2)+\delta(-3)$
$= 1+0+0+0$
$= 1$
$x(1) = 1$





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St. Index = 0+0 = 0
End Index = 3+2 = 5
$L_y = L_x + L_h - 1 = 4+3-1 = 6$

Index	-4	-3	-2	-1	0	1	2	3	4	5	6
x(k)					1	1	1	1			
h(k)					1	2	4				
h(0-k)			4	2	1						
h(1-k)				4	2	1					
h(2-k)					4	2	1				
h(3-k)						4	2	1			
h(4-k)							4	2	1		
h(5-k)								4	2	1	

At n=0,
$y(0) = \sum_{k=-\infty}^{\infty} x(k)\,h(-k)$
$y(0) = 1\times1 = 1$

At n=1,
$y(1) = \sum_{k=-\infty}^{\infty} x(k)\,h(1-k)$
$\therefore y(1) = 2\times1+1\times1 = 3$

At n=2,
$y(2) = \sum_{k=-\infty}^{\infty} x(k)\,h(2-k)$
$y(2) = 4\times1+2\times1+1\times1 = 7$

At n=3,
$y(3) = \sum_{k=-\infty}^{\infty} x(k)\,h(3-k)$
$y(3) = 4\times1+2\times2+1\times1 = 7$

At n=4,
$y(4) = \sum_{k=-\infty}^{\infty} x(k)\,h(4-k)$
$y(4) = 4\times1+2\times1$
$= 6$

At n=5,
$y(5) = \sum_{k=-\infty}^{\infty} x(k)\,h(5-k)$
$y(5) = 4\times1$
$= 4$

Hence, $y(n) = \{1,3,7,7,6,4\}$ Ans

Tabular Method

$x(n) = \{1,1,1,1\}$
$h(n) = \{1,2,4\}$

h(n) \ x(n)	1	1	1	1
1	1	1	1	1
2	2	2	2	2
4	4	4	4	4

[Add elements in each diagonal]

So, $y(n) = \{1,3,7,7,6,4\}$ Ans

Matrix Method

zero padding → Process of adding 0 to sequence

Q) $x(n) = \{4,2,1,3\}$
$h(n) = \{1,2,2,1\}$
↑
Soln

starting index = 0-1 = -1
Ending index = 3+2 = 5
$L_y = L_x + L_h - 1 = 4+4-1 = 7$

Now, After zero padding,
$x(n) = \{4,2,1,3,0,0,0\}$
$h(n) = \{1,2,2,1,0,0,0\}$
↑

PTO →

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$$\begin{bmatrix} 4 & 0 & 0 & 0 & 3 & 1 & 2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 1 \\ 1 & 2 & 4 & 0 & 0 & 0 & 3 \\ 3 & 1 & 2 & 4 & 0 & 0 & 0 \\ 0 & 3 & 1 & 2 & 4 & 0 & 0 \\ 0 & 0 & 3 & 1 & 2 & 4 & 0 \\ 0 & 0 & 0 & 3 & 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2+8 \\ 1+4+8 \\ 3+2+4+4 \\ 6+2+2 \\ 6+1 \\ 3+ \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 13 \\ 13 \\ 10 \\ 7 \\ 3 \end{bmatrix}$$

$\therefore y(n) = \{4, 10, 13, 13, 10, 7, 3\}$ Ans

Q) Find linear convolution using graphical method of signal,

$x(n) = \sum_{k=0}^{3} \delta(n-k)$ $\quad h(n) = 2^n [u(n) - u(n-3)]$

Sol^n

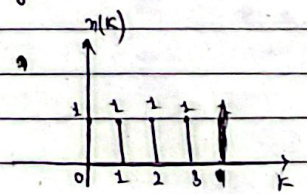
$x(n) = \{1, 1, 1, 1\}$

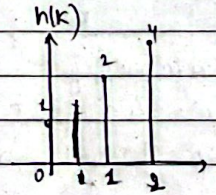
$h(n) = \{1, 2, 4\}$

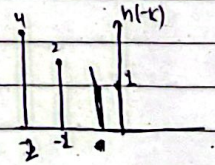
St. Index = 0

End Index = 5

$L_y = L_x + L_h - 1 = 4+3-1 = 6$







$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

For n=0,

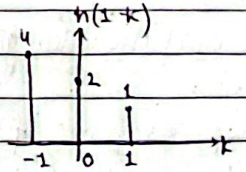
$$y(0) = \sum_{k=-\infty}^{\infty} x(k)\, h(-k)$$

$y(0) = 1 \times 1 = 1$

For n=1,

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)\, h(1-k)$$

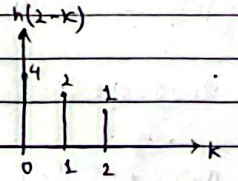
$y(1) = 2 \times 1 + 1 \times 1 = 3$



For n=2,

$$y(2) = \sum_{k=-\infty}^{\infty} x(k)\, h(2-k)$$

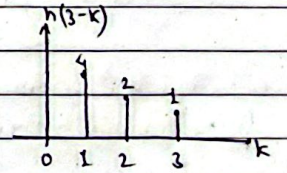
$y(2) = 4 \times 1 + 2 \times 1 + 1 \times 1 = 7$



For n=3,

$$y(3) = \sum_{k=-\infty}^{\infty} x(k)\, h(3-k)$$

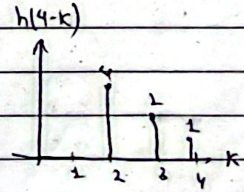
$y(3) = 4 \times 1 + 2 \times 1 + 1 \times 1 = 7$



For n=4,

$$y(4) = \sum_{k=-\infty}^{\infty} x(k)\, h(4-k)$$

$y(4) = 4 \times 1 + 2 \times 1 = 6$



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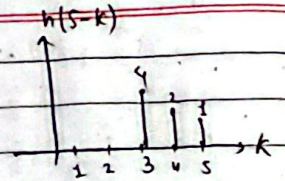
For n=5,

$$y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k)$$



$y(5) = 1 \times 4 = 4$

Hence, $y(n) = \{1, 3, 7, 7, 6, 4\}$ Ans

Q) If the o/p of the system given by $y(n) = \{2, 5, 1, -10, -10, -3, 6, 9\}$ and impulse response of the system $h(n) = \{1, 2, 0, -3\}$. Determine the i/p signal of that system.

Sol:

$y(n) = \{2, 5, 1, -10, -10, -3, 6, 9\}$

$h(n) = \{1, 2, 0, -3\}$ Here, $L_y = 8$, $L_h = 4$

We have, $L_y = L_x + L_h - 1$

or, $8 = L_x + 4 - 1$

$\therefore L_x = 5$

Let, a, b, c, d, e be the element of $x(n)$.

Using tabular method for convolution,

h(n) \ x(n)	a	b	c	d	e
1	a	b	c	d	e
2	2a	2b	2c	2d	2e
0	0	0	0	0	0
-3	-3a	-3b	-3c	-3d	-3e

$y(n) = \{a, 2a+b, 2b+c, -3a+2c+d, -3a+2d+e, -3c+2e, -3d, -3e\}$

Index $y_i = x_i + h_i$

or, $-3 = x_i + 0$

$\therefore x_i = -3$

So, $a = 2$, $-3c = 9$, $-3d = 6$, $2a+b = 5$, $2b+c = 1$,

$\therefore c = -3$ $\therefore d = -2$ $\therefore b = 1$ $\therefore c = -1$

$-3a + 2c + d = -10$

or, $-6 + 2c - 2 = -10$

$\therefore c = -1$

Hence, $x(n) = \{2, 1, -1, -2, -3\}$ Ans

Q) Find the convolution sum $x(n) = u(n)$ and $h(n) = a^n u(n)$.

Sol:

$x(n) = \{1, 1, 1, \ldots\}$ $x(k) = u(k)$

$h(n) = \{a^1, \ldots\}$ $h(k) = a^n u(k)$

$h(n-k) = a^{n-k} u(n-k)$

For convolution,

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} u(k)\, a^{n-k} u(n-k)$$

$$= \sum_{k=0}^{n} a^{n-k}$$

$$= a^n \sum_{k=0}^{n} a^{-k}$$

$$= a^n \left[\frac{1-(a^{-1})^{n+1}}{1-(a^{-1})}\right]$$

$$= a^n \left(\frac{1-\frac{1}{a^{n+1}}}{1-\frac{1}{a}}\right)$$

$$= a^n \left(\frac{a^{n+1}-1}{a^n.a} \times \frac{a}{a-1}\right)$$

$$= \frac{a^{n+1}-1}{a-1}$$ Ans

($\frac{a(1-r^{n+1})}{1-r}$)

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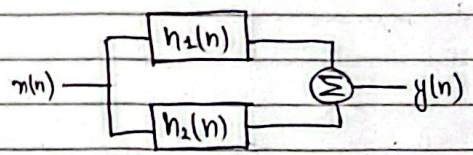
2013 Spring

Q) Two sub system $h_1(n)$ and $h_2(n)$ are interconnected as shown in the block diagram. Determine the response of the system is $h_1(n) = \{1, 4, 2\}$ and $h_2(n) = \{2, 3, 1\}$ when excited by input $x(n) = \{2, 4\}$.



Soln

Let the o/p of subsystems 1 and 2 be $y_1(n)$ and $y_2(n)$ respectively.

Here, $y_1(n) = x(n) * h_1(n)$ —(i)

$y_2(n) = x(n) * h_2(n)$ —(ii)

Also, $y(n) = y_1(n) + y_2(n)$ —(iii)

For eqⁿ (i), Using tabular method to find convolution,

$h_1(n)$ \ $x(n)$	2	4
1	2	4
4	8	16
2	4	8

So, $y_1(n) = \{2, 12, 20, 8\}$

For eqⁿ (ii), Using tabular method to find convolution,

$h_2(n)$ \ $x(n)$	2	4
2	4	8
3	6	12
1	2	4

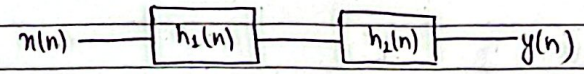
So, $y_2(n) = \{4, 14, 14, 4\}$

From eqⁿ (iii), $y(n) = y_1(n) + y_2(n)$

$= \{2, 12, 20, 8\} + \{4, 14, 14, 4\}$

$y(n) = \{6, 26, 34, 12\}$ Ans

Q) Two - - - (same as above)



Soln

Let the o/p of subsystems 1 and 2 be $y_1(n)$ & $y(n)$ respectively.

Here, $y_1(n) = x(n) * h_1(n)$ —(i)

$y(n) = y_1(n) * h_2(n)$ —(ii)

For eqⁿ (i), Using tabular method,

$x(n)$ \ $h_1(n)$	1	4	2
2	2	8	4
4	4	16	8

$y_1(n) = \{2, 12, 20, 8\}$

For eqⁿ (ii), Using tabular method,

$y_1(n)$ \ $h_2(n)$	2	3	1
2	4	6	2
12	24	36	12
20	40	60	20
8	16	24	8

$y(n) = \{4, 30, 78, 88, 44, 8\}$ Ans

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Q) Find the convolution sum of the signal $x(n)=\{1,-1,1,-1\}$ and $h(n)=\{1,2,3,4\}$ using graphical method.

Sol^n

$x(n)=\{1,-1,1,-1\}$

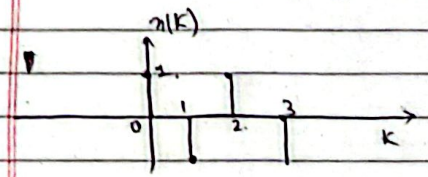
$h(n)=\{1,2,3,4\}$

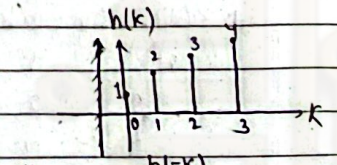
Starting Index = 0+0 = 0

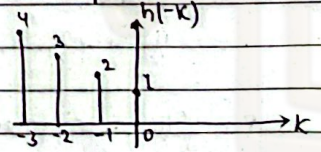
Ending Index = 3+3 = 6

$L_y = L_x + L_h - 1$ or, $L_y = 4+4-1 = 7$

Now, $x(k)=\{1,-1,1,-1\}$, $h(k)=\{1,2,3,4\}$







We know,

$$y(n)=\sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$$

For n=0,

$$y(0)=\sum_{k=-\infty}^{\infty} x(k)\,h(-k)$$

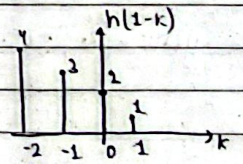
$y(0) = 1\times1 = 1$

For n=1,

$$y(1)=\sum_{k=-\infty}^{\infty} x(k)\,h(1-k)$$

$y(1) = 1\times2+(-1)\times1$

$= 1$

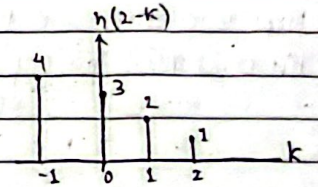


For n=2,

$$y(2)=\sum_{k=-\infty}^{\infty} x(k)\,h(2-k)$$

$y(2) = 1\times3+2\times(-1)+1\times1$

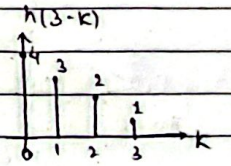
$= 2$



For n=3,

$$y(3)=\sum_{k=-\infty}^{\infty} x(k)\,h(3-k)$$

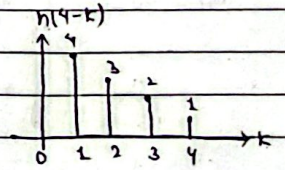
$y(3) = 1\times4+3\times(-1)+2\times1+1\times(-1) = 2$



For n=4,

$$y(4)=\sum_{k=-\infty}^{\infty} x(k)\,h(4-k)$$

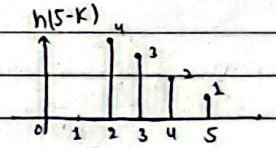
$y(4) = 4\times(-1)+3\times1+2\times(-1) = -3$



For n=5,

$$y(5)=\sum_{k=-\infty}^{\infty} x(k)\,h(5-k)$$

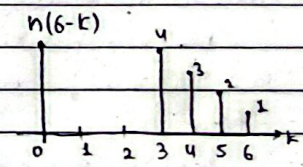
$y(5) = 4\times1+3\times(-1) = 1$



For n=6,

$$y(6)=\sum_{k=-\infty}^{\infty} x(k)\,h(6-k)$$

$y(6) = 4\times(-1) = -4$



Hence, $y(n)=\{1,1,2,2,-3,1,-4\}$ Ans

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Discrete time Fourier series/Fourier transform properties with proof

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Fourier series and its properties

The exponential form of discrete series can be expressed as,

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\omega_0 kn} \quad \text{for all } n$$

where, $X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\omega_0 kn}$ for all k

The trigonometric form of fourier series

$$x(t) = a_0 + \sum_{n=0}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

where, $a_0 = \frac{1}{T}\int_0^T x(t)\,dt$

$$a_n = \frac{2}{T}\int_0^T x(t)\cos n\omega_0 t\, dt \quad , n = 1, 2, \ldots$$

$$b_n = \frac{2}{T}\int_0^T x(t)\sin n\omega_0 t\, dt \quad , n = 1, 2, \ldots$$

Properties of Discrete Fourier Series

1) Linearity

Consider two periodic sequences $x_1(n)$ and $x_2(n)$ both with period N, such that

$DFS[x_1(n)] = X_1(k)$

$DFS[x_2(n)] = X_2(k)$

then,

$DFS[a_1x_1(n) + a_2x_2(n)] = a_1X_1(k) + a_2X_2(k)$

2) x

$\omega_0 = \frac{2\pi}{N}$

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2) Time shifting

If $x(n)$ is a periodic sequence with N samples and $DFS[x(n)] = X(k)$

then, $DFS[x(n-m)] = e^{-j\omega_0 km} X(k)$

where,

$(n-m)$ is the shifted version of $x(n)$.

3) Symmetry Property / conjugation Property

we have, $DFS[x(n)] = X(k)$

Now, $DFS[x^*(n)] = X^*(-k)$ and

$DFS[x^*(-n)] = X^*(k)$

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4) Periodic Convolution

Let $x_1(n)$ and $x_2(n)$ be two periodic sequences with period N with

$DFS[x_1(n)] = X_1(k)$

and $DFS[x_2(n)] = X_2(k)$

If $X_3(k) = X_1(k).X_2(k)$, then the periodic sequence $x_3(n)$ with fourier series coefficient $X_3(k)$ is,

$$x_3(n) = \sum_{m=0}^{N-1} x_1(m)\, x_2(n-m)$$

Hence, $DFS[x_1(m)\, x_2(n-m)] = X_1(k)X_2(k)$

$n \rightarrow +ve \rightarrow$ right shift $\rightarrow$ anticlockwise (in circular)

$n \rightarrow -ve \rightarrow$ left shift $\rightarrow$ clockwise

~~Q) Find the convolution bet^n two signals $x(n) = u(n)$ & $h(n) = a^n u(n)$~~

Circular Convolution

$$y(n) = x(n) \textcircled{N} h(n) = \sum_{k=0}^{N-1} x(k)h(n-k)$$

Graphical Method

Q) $x_1(n) = \{1,2,3,4\}$

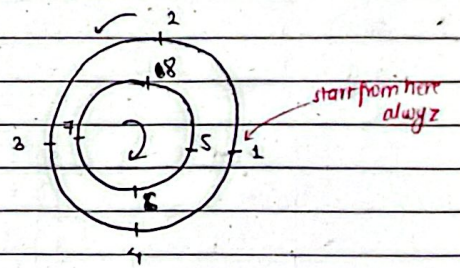
$x_2(n) = \{5,6,7,8\}$

For n = 0,

$$y(0) = \sum_{k=0}^{3} x(k)h(-k)$$

$= 1\times5 + 8\times8 + 3\times7 + 6\times4$

$= 66$

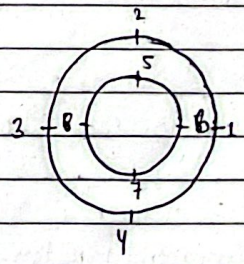


For n = 1,

$$y(1) = \sum_{k=0}^{3} x(k)h(1-k)$$

$= 1\times6 + 5\times2 + 3\times8 + 7\times4$

$= 68$

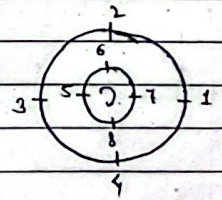


For n = 2,

$$y(2) = \sum_{k=0}^{3} x(k)h(2-k)$$

$= 7\times1 + 2\times6 + 3\times5 + 8\times4$

$= 66$



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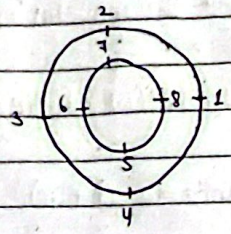
For n=3,

$$y(3) = \sum_{k=0}^{3} x(k)\,h(3-k)$$

$$= 8\times1 + 7\times2 + 3\times6 + 5\times4$$

$$= 60$$



$y(n) = \{70, 68, 66, 60\}$ Ans

Matrix Method

$x_1(n) = \{1,2,3,4\}$

$x_2(n) = \{5,6,7,8\}$

Ans

$$\begin{bmatrix} 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 66 \\ 68 \\ 66 \\ 60 \end{bmatrix}$$ Ans

a) Find the circular convolution of two sequences $x_1(n) = \{1,2,-1,-2,3,1\}$ $x_2(n) = \{3,2,1\}$

Using matrix method, Adding zero padding, $x_2(n) = \{3,2,1,0,0,0\}$

$$\begin{bmatrix} 1 & 1 & 3 & -2 & -1 & 2 \\ 2 & 1 & 1 & 3 & -2 & -1 \\ -1 & 2 & 1 & 1 & 3 & -2 \\ -2 & -1 & 2 & 1 & 1 & 3 \\ 3 & -2 & -1 & 2 & 1 & 1 \\ 1 & 3 & -2 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \\ 2 \\ -6 \\ 4 \\ 7 \end{bmatrix}$$

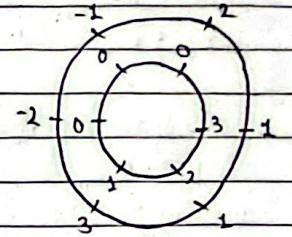
Using Graphical method,

For n=0,

$$y(0) = \sum_{k=0}^{5} x(k)\,h(-k)$$

$$= 3\times1 + 2\times1 + 3\times1$$

$$= 8$$

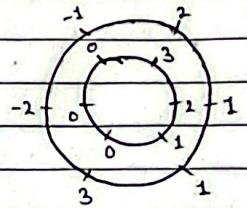


For n=1,

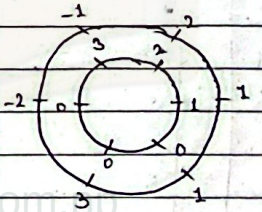
$$y(1) = \sum_{k=0}^{5} x(k)\,h(1-k)$$

$$= 3\times2 + 2\times1 + 1\times1$$

$$= 6+2+1 = 9$$



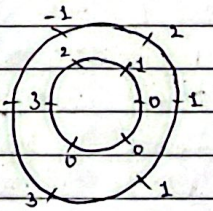
For n=2,



$y(2) = 1+4-3$

$= 2$

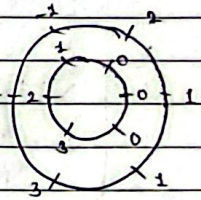
For n=3,



$y(3) = -6-2+2$

$= -6$

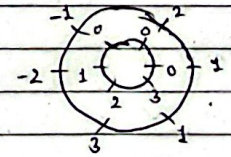
For n=4,



$y(4) = 9-4-1$

$= 4$

For n=5,



$y(5) = -2+6+3$

$= 7$ Ans

Hence, $y(n) = \{8,9,2,-6,4,7\}$ Ans

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Linear convolution using circular convolution

$x_1(n) = \{1,2,3,4\}$
$x_2(n) = \{5,6,7,8\}$

Soln

The no. of element of output y(n) of linear convolution is given by,

$L_y = L_x + L_h - 1 = 4+4-1 = 7$

∴ For using circular convolution, the no. of elements of each sequences must be equal to 7.

This is attained by padding zeros.

∴ $x_1(n) = \{1,2,3,4,0,0,0\}$
$x_2(n) = \{5,6,7,8,0,0,0\}$

Using Matrix method,

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 4 & 3 & 2 \\ 2 & 1 & 0 & 0 & 0 & 4 & 3 \\ 3 & 2 & 1 & 0 & 0 & 0 & 4 \\ 4 & 3 & 2 & 1 & 0 & 0 & 0 \\ 0 & 4 & 3 & 2 & 1 & 0 & 0 \\ 0 & 0 & 4 & 3 & 2 & 1 & 0 \\ 0 & 0 & 0 & 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \\ 34 \\ 60 \\ 61 \\ 52 \\ 32 \end{bmatrix}$$

Hence, $y(n) = \{5,11,34,60,61,52,32\}$ Ans

circular convolution using Linear convolution

$x_1(n) = \{1,2,3,4\}$
$x_2(n) = \{5,6,7,8\}$

Soln

The no. of element of output y(n) of linear convolution is given by,

$L_y = L_x + L_h - 1 = 4+4-1 = 7$

Finding linear convolution,

$x_1(n)$ \ $x_2(n)$	5	6	7	8
1	5	6	7	8
2	10	12	14	16
3	15	18	21	24
4	20	24	28	32

From linear convolution, $y(n) = \{5,16,34,60,61,52,32\}$

Now, For circular convolution,

K (Index)	0	1	2	3
	5	16	34	60
	61	52	32	0
	66	68	66	60

Hence, circular convolution is, $y(n) = \{66,68,66,60\}$ Ans

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Discrete Fourier series (DFS)

Definition:

For a periodic discrete-time signal $x[n]$ with period N, the DFS represents $x[n]$ as a linear combination of complex exponentials.

$$x[n] = \sum_{K=0}^{N-1} x[K]\, e^{j2\pi Kn/N}, \quad n = 0,1,2,\ldots N-1$$

where the DFS coefficients $x[K]$ are

$$x[K] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-j2\pi Kn/N}, \quad K = 0,1,2,\ldots N-1$$

Here, N → period of the sequence

K → frequency index

Properties:

1. Linearity

If $x_1[n] \longleftrightarrow x_1[K]$ & $x_2[n] \longleftrightarrow x_2[K]$, then

$$ax_1[n] + bx_2[n] \longleftrightarrow ax_1[K] + bx_2[K]$$

Proof:

$$x[K] = \frac{1}{N}\sum_{n=0}^{N-1} (ax_1[n] + bx_2[n])\, e^{-j2\pi Kn/N}$$

$$= a\cdot\frac{1}{N}\sum_{n=0}^{N-1} x_1[n]\, e^{-j2\pi Kn/N} + b\frac{1}{N}\sum_{n=0}^{N-1} x_2[n]\, e^{-j2\pi Kn/N}$$

So, linear combination in time domain corresponds to same combination in frequency domain.

2. Time shifting

If $[(n-n_0) \bmod N] \longleftrightarrow x[K]$, then

$$x[(n-n_0)] \longleftrightarrow x[K]\, e^{-j2\pi Kn_0/N}$$

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Proof:

$$X_{shift}[K] = \frac{1}{N}\sum_{n=0}^{N-1} x[(n-n_0)]\, e^{-j2\pi Kn/N}$$

Let $m = n - n_0 \Rightarrow n = m + n_0$

$$X_{shift}[K] = \frac{1}{N}\sum_{m=0}^{N-1} x[m]\, e^{-j2\pi K(m+n_0)/N}$$

$$= e^{-j2\pi Kn_0/N}\cdot\frac{1}{N}\sum_{m=0}^{N-1} x[m]\, e^{-j2\pi Km/N}$$

$$= X[K]\, e^{-j2\pi Kn_0/N}$$

So, time shift introduces a linear phase factor in DFS coefficients.

3. Frequency shifting

If $x[n]\, e^{j2\pi K_0 n/N} \longleftrightarrow X[(K-K_0) \bmod N]$

Proof: $X_{shift}[K] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{j2\pi K_0 n/N}\cdot e^{-j2\pi Kn/N}$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-j2\pi(K-K_0)n/N}$$

$$= X[(K-K_0) \bmod N]$$

So, frequency shift corresponds to circular shift of DFS coefficients.

4. Conjugation

If $x^*[n] \longleftrightarrow X^*[-K \bmod N]$

Proof:

$$X_{conj}[K] = \frac{1}{N}\sum_{n=0}^{N-1} x^*[n]\, e^{-j2\pi Kn/N}$$

$$= \left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{j2\pi Kn/N}\right)^*$$

$$= X^*[-K \bmod N]$$

So, conjugating a sequence flips & conjugates its DFS coefficients.

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5. Parseval's Theorem

$$\sum_{n=0}^{N-1} |x[n]|^2 = N \sum_{k=0}^{N-1} |x[k]|^2$$

Proof: $x[n] = \sum_{k=0}^{N-1} x[k]\, e^{j2\pi kn/N}$

$$\sum_{n=0}^{N-1} |x[n]|^2 = \sum_{n=0}^{N-1} \left(\sum_{k=0}^{N-1} x[k]\, e^{j2\pi kn/N} \right) \left(\sum_{l=0}^{N-1} x^*[l]\, e^{-j2\pi ln/N} \right)$$

$$= \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} x[k]\, x^*[l] \sum_{n=0}^{N-1} e^{j2\pi(k-l)n/N}$$

$$= \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} x[k]\, x^*[l]\, N \cdot \delta[k-l]$$

$$= N \sum_{k=0}^{N-1} |x[k]|^2$$

This shows energy in time ≈ scaled energy in frequency.

Discrete Time Fourier Transform [DTFT]

For aperiodic sequence $x[n]$:

$$x[e^{j\omega}] = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Inverse, $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} x[e^{j\omega}]\, e^{j\omega n}\, d\omega$

Properties:

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$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$.

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Z-Transform

The Z-transform of discrete time signal $x(n)$ is defined by:

$$x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Region of Convergence (ROC)

The set of values of z or equivalently the set of points on z-plane for which x of z converges is called ROC.

Properties

(i) The ROC of z-transform is a ring or disc in the z-plane centered at the origin.

(ii) The ROC of the z-transform cannot contain any poles.

(iii) The ROC of z-transform of an LTI system contains the unit circle.

(iv) If $x(n) = \delta(n)$ i.e. impulse response is the only sequence whose ROC of z-transform is the entire z-plane.

(v) If $x(n)$ is the infinite duration causal sequence then its ROC is

$|z| > a$

i.e. it is the exterior of a circle of radius = a.

(vi) If $x(n)$ is an infinite duration anti-causal sequence then its ROC is $|z| < b$ ie. it is the interior of the circle of radius = b.

Q) Find the z-transform of $x(n) = a^n u(n)$.

Sol^n

$$x(z) = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$\left[\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}\right]$ $\cos n = \frac{e^{jn} + e^{-jn}}{2}$

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$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{1}{1-(az^{-1})} = \frac{z}{z-a}$$ Ans

Q) Find the z-transform of $x(n) = \left(\frac{1}{4}\right)^n \cos\left(\frac{n\pi}{3}\right) u(n)$

Sol^n

$$x(z) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{4}\right)^n \cos\left(\frac{n\pi}{3}\right) u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n \cos\left(\frac{n\pi}{3}\right) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n \cdot \left[\frac{e^{j\pi n/3} + e^{-j\pi n/3}}{2}\right] z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{1}{4}\right)^n e^{j\pi n/3} z^{-n} + \sum_{n=0}^{\infty} \frac{1}{2} \cdot \left(\frac{1}{4}\right)^n e^{-j\pi n/3} z^{-n}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{4} e^{j\pi/3} z^{-1}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{4} e^{-j\frac{\pi}{3}} z^{-1}\right)^n$$

$$= \frac{1}{2} \times \frac{1}{\left(1 - \frac{1}{4} e^{j\pi/3} z^{-1}\right)} + \frac{1}{2} \times \frac{1}{1 - \left(\frac{1}{4} e^{-j\pi/3} z^{-1}\right)}$$

$$= \frac{1}{2} \times \frac{2 \cdot 4z}{4z - e^{j\pi/3}} + \frac{1}{2} \times \frac{4z}{4z - e^{-j\pi/3}}$$

$$= \frac{2z}{4z - e^{j\pi/3}} + \frac{2z}{4z - e^{-j\pi/3}}$$

$$= 2z \left(\frac{4z - e^{-j\pi/3} + 4z - e^{j\pi/3}}{16z^2 - 4ze^{-j\pi/3} - 4ze^{j\pi/3} + e^{0}}\right)$$

$$= 2z \left(\frac{8z - \left(e^{j\pi/3} + e^{-j\pi/3}\right)}{16z^2 + 1 - 4z\left(e^{-j\pi/3} + e^{j\pi/3}\right)}\right)$$

$$= 2z \left(\frac{8z - 2 \times \cos\frac{\pi}{3}}{16z^2 + 1 - 4z \times 2 \times \cos\frac{\pi}{3}}\right)$$

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→ ~~If Poles & zeros of any system are within unit circle...~~

stable → Poles lie strictly inside unit circle.
→ ROC includes unit circle

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$$= 2z \cdot \left(\frac{8z-1}{16z^2+1-4z}\right)$$

$$= \frac{z(16z-2)}{16z^2-4z+1}$$

$$= \frac{z(16z-2)}{(z-(0.125+0.216j))(z-(0.125-0.216j))}$$

Poles = $0.125+0.216j$, $0.125-0.216j$

Zeros = $0, \frac{1}{8}$ Ans

Q) Find z-transform of $x(n) = a^n u(n-1)$.

Soln

$$x(n) = \begin{cases} +a^n & , n \le -1 \\ 0 & , n \ge 0 \end{cases} \quad \text{because} \quad u(-n-1) = \begin{cases} 1 \text{ for } n \le -1 \\ 0 \text{ for } n \ge 0 \end{cases}$$

$$x(z) = \sum_{n=-\infty}^{\infty} a^n u(n-1) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= \sum_{n=-\infty}^{-1} a^n \left[\frac{1}{z^n}\right]^n$$

$$= \sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= a^n \times \frac{1}{1-\frac{a}{z}}$$

$$= \sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= \sum_{n=1}^{\infty} (z^n a^{-n})$$

$$= a^n \times \left[\frac{1}{1+za} \sum_{n=1}^{\infty} (za)^n\right]$$

$$= \frac{a^n}{1+z} - a^n$$

$$= a^n \left(\frac{1}{1-z} - 1\right)$$

$$= a$$

$$= \sum_{n=0}^{\infty} \left[(a^{-1}z)^n - 1\right]$$

$$= \sum_{n=0}^{\infty} (a^{-1}z)^n - \sum_{n=0}^{\infty} (1)$$

$$= \frac{1}{1-a^{-1}z} - 1$$

$$= \frac{1}{1-\frac{z}{a}} - 1$$

$$= \frac{a}{a-z} - 1 = \frac{a-a+z}{a-z} = \frac{z}{a-z}$$ Ans

Q) Find the z-transform and ROC of $x(n) = 2\left(\frac{5}{6}\right)^n u(-n-1) + 3\left(\frac{1}{2}\right)^{2n} u(n)$.

Soln

$$x(z) = \sum_{n=-\infty}^{\infty} \left[2\left(\frac{5}{6}\right)^n u(-n-1) + 3\left(\frac{1}{2}\right)^{2n} u(n)\right] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} 2\left(\frac{5}{6}\right)^n u(-n-1) z^{-n} + \sum_{n=-\infty}^{\infty} 3\left(\frac{1}{2}\right)^{2n} u(n) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} 2\left(\frac{5}{6}\right)^n z^{-n} + \sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^{2n} z^{-n}$$

$$= \sum_{n=1}^{\infty} 2\left(\frac{6}{5}\right)^n z^n + \sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^{2n} z^{-n}$$

$$= 2 \times \frac{1}{1-\frac{6}{5}z} + 3 \times \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$= \frac{2\times 6}{6-5z} + \frac{3 \times 2z}{2z-1}$$

$$= -2 \sum_{n=0}^{\infty} \left(\left(\frac{6}{5}\right)^n z^n - 1\right) + 3 \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$$

$$= -2 + 2 \sum_{n=0}^{\infty} \left(\frac{6z}{5}\right)^n + 3 \sum_{n=0}^{\infty} \left(\frac{1}{4z}\right)^n$$

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$$= -2 + 2\times\frac{1}{1-\frac{6z}{5}} + 3\times\frac{1}{1-\frac{1}{4z}}$$

$$= -2 + \frac{10}{5-6z} + \frac{12z}{4z-1}$$

$$= -2 + \frac{40z-10+60z-72z}{20z-5-24z^2+6z}$$

$$= \frac{48z^2-52z+10+28z-10}{-24z^2+26z-5}$$

$$= \frac{z(48z-24)}{-24z^2+26z-5}$$

$$= \frac{z(48z-24)}{(z-0.25)(z-0.833)}$$

zeros = 0, 0.5

Poles = 0.25, 0.833 Ans

Properties of z-transform

1. Linearity

If $x_1(n) \xleftrightarrow{ZT} x_1(z)$ with ROC = R_1

& $x_2(n) \xleftrightarrow{ZT} x_2(z)$ with ROC = R_2

then,

$ax_1(n) + bx_2(n) \xleftrightarrow{ZT} ax_1(z) + bx_2(z)$ with ROC = $R_1 \cap R_2$

Proof:

We know that,

$$Z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$Z[ax_1(n)+bx_2(n)] = \sum_{n=-\infty}^{\infty} [ax_1(n)+bx_2(n)]z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} ax_1(n)z^{-n} + \sum_{n=-\infty}^{\infty} bx_2(n)z^{-n}$$

$$= ax_1(z) + bx_2(z)$$

2. Time shift Property

If $x(n) \xleftrightarrow{ZT} x(z)$ with ROC = R

then, $x(n-m) \xleftrightarrow{ZT} z^{-m}x(z)$ with ROC = R

Proof:

We know,

$$Z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$Z[x(n-m)] = \sum_{n=-\infty}^{\infty} x(n-m)z^{-n}$$

Let, $n-m = p$

then,

$$Z[x(n-m)] = \sum_{p=-\infty}^{\infty} x(p)z^{-(p+m)}$$

$$= \sum_{p=-\infty}^{\infty} x(p)z^{-p}z^{-m}$$

$$= z^{-m}x(z)$$

3. Time Reversal Property

If $x(n) \xleftrightarrow{ZT} x(z)$ with ROC = R

then, $x(-n) \xleftrightarrow{ZT} x\left(\frac{1}{z}\right)$ with ROC = $\frac{1}{R}$

AND

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Proof:

We know that,

$$Z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Z[x(-n)] = \sum_{n=-\infty}^{\infty} x(-n) z^{-n}$$

Let $p = -n$

then,

$$Z[x(-n)] = \sum_{p=-\infty}^{\infty} x(p) z^{p}$$

$$= \sum_{p=-\infty}^{\infty} x(p) (z^{-1})^{-p}$$

$$= x(z^{-1})$$

$$= x\left(\frac{1}{z}\right)$$

4. Multiplication by exponential sequence property

If $x(n) \xleftrightarrow{ZT} x(z)$ with ROC = R

then, $a^n x(n) \xleftrightarrow{ZT} x\left(\frac{z}{a}\right)$ with ROC = |a|R

Proof:

We know that,

$$Z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Z[a^n x(n)] = \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n} = \sum_{n=-\infty}^{\infty} x(n) (a^{-1}z)^{-n}$$

$$= x(a^{-1}z)$$

$$= x\left(\frac{z}{a}\right)$$

5. Convolution Property

If $x_1(n) \xleftrightarrow{ZT} x_1(z)$ with ROC = R_1

& $x_2(n) \xleftrightarrow{ZT} x_2(z)$ with ROC = R_2

then,

$$x_1(n) * x_2(n) \xleftrightarrow{ZT} x_1(z) x_2(z) \text{ with ROC} = R_1 \cap R_2$$

Proof:

We know that,

$$Z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Z[x_1(n) * x_2(n)] = \sum_{n=-\infty}^{\infty} [x_1(n) * x_2(n)] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) \right] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) z^{-(n-k)} z^{-k}$$

$$= \sum_{k=-\infty}^{\infty} x_1(k) z^{-k} \sum_{n=-\infty}^{\infty} x_2(n-k) z^{-(n-k)}$$

Let $n-k = p$

then,

$$Z[x_1(n) * x_2(n)] = \sum_{k=-\infty}^{\infty} x_1(k) z^{-k} \cdot \sum_{p=-\infty}^{\infty} x_2(p) z^{-p}$$

$$= x_1(z) x_2(z)$$

6. Multiplication by n or differentiation property

If $x(n) \xleftrightarrow{ZT} x(z)$ with ROC = R

then, $nx(n) \xleftrightarrow{ZT} -z \frac{dx(z)}{dz}$ with ROC = R

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Proof:

We know that,

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Differentiating w.r.to z, we get,

$$\frac{dX(z)}{dz} = \frac{d}{dz}\left[\sum_{n=-\infty}^{\infty} x(n) z^{-n}\right]$$

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} x(n) \frac{dz^{-n}}{dz}$$

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} x(n) x(-n) \times z^{-n-1}$$

$$\frac{dX(z)}{dz} = -z^{-1} \sum_{n=-\infty}^{\infty} n x(n) z^{-n}$$

$$-z \frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} n x(n) z^{-n}$$

This implies: $n x(n) \xleftrightarrow{ZT} -z \frac{dX(z)}{dz}$

Inf 7. Multiplication Property

If $x_1(n) \xleftrightarrow{ZT} X_1(z)$ with ROC = R_1

& $x_2(n) \xleftrightarrow{ZT} X_2(z)$ with ROC = R_2

then,

$$x_1(n) x_2(n) \xleftrightarrow{ZT} \frac{1}{2\pi j} \oint X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv$$

PTO

Proof:

We know that,

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Z[x_1(n) x_2(n)] = \sum_{n=-\infty}^{\infty} [x_1(n) x_2(n)] z^{-n}$$

But from the definition of inverse z-transform,

$$x(n) = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

Now, changing the variable z by v & substituting, we get,

$$Z[x_1(n) x_2(n)] = \sum_{n=-\infty}^{\infty} \frac{1}{2\pi j} \oint X_1(v) v^{n-1} dv \; x_1(n) z^{-n}$$

$$= \frac{1}{2\pi j} \oint X_1(v) v^{-1} dv \sum_{n=-\infty}^{\infty} x_2(n) \left(\frac{z}{v}\right)^{-n}$$

$$= \frac{1}{2\pi j} \oint X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv$$

$\left[\because x_1(n) = \frac{1}{2\pi j} \oint X_1(v) v^{n-1} dv\right]$.

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Q) Find z-transform of

(i) $x(n) = a^n u(n-2)$

Sol^n

$$Z[u(n)] = \sum_{n=-\infty}^{\infty} u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} 1 (z^{-1})^n$$

$$= \frac{1}{1-z^{-1}} = \frac{z}{z-1}$$

Using time shifting theorem,

$$Z[u(n-2)] = z^{-2} \times \frac{z}{z-1} = \frac{z^{-1}}{z-1}$$

Using multiplication by exponential sequence property,

$$Z[a^n u(n-2)] = \left(\frac{z^{-1}}{z-1}\right)_{z \to \frac{z}{a}}$$

$$= \frac{\left(\frac{z}{a}\right)^{-1}}{\frac{z}{a} - 1}$$

$$= \frac{a}{z} \times \frac{a}{z-a} = \frac{a^2}{z(z-a)}$$ Ans

(ii) $x(n) = -a^n u(-n-1)$

Sol^n

$$Z[u(n)] = \frac{z}{z-1}$$

Using time reversal property,

$$Z[u(-n)] = \left(\frac{z}{z-1}\right)_{z \to \frac{1}{z}}$$

$$= \left(\frac{1/z}{1/z - 1}\right) = \frac{1}{z} \times \frac{z}{1-z} = \frac{1}{1-z}$$

Using time shifting theorem,

$$Z[u(-n-1)] = z^{-1} \times \frac{1}{1-z} = \frac{1}{z(1-z)}$$

Using multiplication by exponential sequence property,

$$Z[-a^n u(-n-1)] = -Z[a^n u(-n-1)]$$

$$= -1 \times \left[\frac{1}{z(1-z)}\right]_{z \to \frac{z}{a}}$$

$$= -1 \times \left[\frac{1}{\frac{z}{a}\left(1-\frac{z}{a}\right)}\right]$$

$$= -1 \times \frac{a^2}{z \times (a-z)} = \frac{a^2}{z(z-a)}$$ Ans

2015 fall

Q) Find the z-transform of:

(i) $x(n) = \left[\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n\right] u(n-5)$

Sol^n

$$x(n) = \left(\frac{1}{2}\right)^n u(n-5) + \left(\frac{3}{4}\right)^n u(n-5)$$

$$x(n) = x_1(n) + x_2(n)$$

where, $x_1(n) = \left(\frac{1}{2}\right)^n u(n-5)$

$$x_2(n) = \left(\frac{3}{4}\right)^n u(n-5)$$

For $Z[x_1(n)]$,

$$Z[u(n)] = \frac{z}{z-1}$$

Using time shifting theorem,

$$Z[u(n-5)] = z^{-5} \times \frac{z}{z-1} = \frac{z^{-4}}{z-1}$$

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Using multiplication by exponential sequence property,

$$Z\left[\left(\frac{1}{2}\right)^n u(n-5)\right] = \left(\frac{z^{-4}}{z-1}\right)_{z \to \frac{z}{1/2}}$$

$$= \left(\frac{z^{-4}}{z-1}\right)_{z \to 2z}$$

$$= \frac{(2z)^{-4}}{2z-1}$$

$$= \frac{1}{16z^4(2z-1)}$$

Similarly, for $x_2(z)$,

Using multiplication by exponential sequence property.

$$Z\left[\left(\frac{3}{4}\right)^n u(n-5)\right] = \left(\frac{z^{-4}}{z-1}\right)_{z \to \frac{z}{3/4}}$$

$$= \left(\frac{z^{-4}}{z-1}\right)_{z \to 4z/3}$$

$$= \left(\frac{(4z/3)^{-4}}{\frac{4z}{3}-1}\right)$$

$$= \left(\frac{3}{4z}\right)^4 \times \frac{3}{4z-3}$$

$$= \frac{243}{256z^4(4z-3)}$$

We have, $x(z) = x_1(z) + x_2(z)$

$$= \frac{1}{16z^4(2z-1)} + \frac{243}{256z^4(4z-3)}$$

$$= \frac{16(4z-3) + 243(2z-1)}{256z^4(2z-1)(4z-3)}$$

$$= \frac{64z - 48 + 486z - 243}{256z^4(2z-1)(4z-3)}$$

$$= \frac{550z - 251}{256z^4(2z-1)(4z-3)}$$ Ans

Poles: $0, \frac{1}{2}, \frac{3}{4}$

ROC: $0 < z < \frac{3}{4}$

(ii) $x(n) = \left(\frac{1}{2}\right)^n u(n) + (1+y)^n u(-n-1)$

Sol^n

$$Z[u(n)] = \frac{z}{z-1} \quad \text{— (i)}$$

~~Using (i)~~

Using multiplication by exponential sequence property,

$$Z\left[\left(\frac{1}{2}\right)^n u(n)\right] = \left(\frac{z}{z-1}\right)_{z \to \frac{z}{1/2}}$$

$$= \frac{2z}{2z-1}$$

From eq^n (i),

Using time reversal property,

$$Z[u(-n)] = \left(\frac{z}{z-1}\right)_{z \to \frac{1}{z}}$$

$$= \frac{1/z}{1/z - 1} = \frac{1}{z} \times \frac{z}{1-z} = \frac{1}{1-z}$$

Using time shifting theorem,

$$Z[u(-n-1)] = z^{-1} \times \frac{1}{1-z} = \frac{1}{z(1-z)}$$

Using multiplicarion by exponential sequence property,

$$Z[(1+y)^n u(-n-1)] = \left[\frac{1}{z(1-z)}\right]_{z \to \frac{z}{1+y}}$$

$$= \frac{1}{\frac{z}{1+y}\left(1 - \frac{z}{1+y}\right)}$$

$$= \frac{1+y}{z} \times \frac{1+y}{1+y-z}$$

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$$= \frac{(1+y)^2}{z+zy-z^2}$$

So, $z[x(n)] = x(z_1) + x(z_2)$

$$= \frac{2z}{2z-1} + \frac{(1+y)^2}{(z+zy-z^2)}$$

$$= \frac{2z^2(1+y-z)+(1+y)^2(2z-1)}{(2z-1)z(1+y-z)}$$

$$= \frac{2z^2+2yz^2-2z^3+2z-1+4yz-2y+2y^2z-y^2}{z(2z-1)(1+y-z)}$$

$$\therefore z[x(n)] = \frac{2z+2z^2-2z^3-2y-y^2+2yz^2+4yz+2y^2z-1}{z(2z-1)(1+y-z)}$$

Ans

Inverse z-transform

It is defined as,

$$x(n) = \frac{1}{2\pi j}\oint X(z) z^{n-1} dz$$

{ ROC > |a| → causal signal
ROC < |a| → non causal signal
|a| → some value }

{ Causal signal मा : z को power ascending order मा हुन्छ ।
Non causal signal मा : z को power descending order मा हुन्छ । }

Two methods:

(i) Long Division Method

(ii) Partial Fraction Method

(i) Long Division Method

Q) Find the inverse z-transform:

$$X(z) = \frac{z^2+2z}{z^3-3z^2+4z+1} \quad ; ROC > |a|$$

Soln

$$z^3-3z^2+4z+1 \;\big|\; z^2+2z \quad (z^{-1}+5z^{-2}+11z^{-3}+12z^{-4}\ldots)$$

$z^2-3z+4+z^{-1}$ (-) (+) (-) (-)

$5z+4+z^{-1}$

$5z-15+20z^{-1}+5z^{-2}$ (-) (+) (-) (-)

$11-19z^{-1}-5z^{-2}$

$11-33z^{-1}+44z^{-2}+11z^{-3}$ (-) (+) (-) (-)

$14z^{-1}-49z^{-2}+11z^{-3}$

$12z^{-1}-36z^{-2}+48z^{-3}+12z^{-4}$ (-) (+) (-) (-)

$-13z^{-2}-59z^{-3}-12z^{-4}$

$X(z) = z^{-1}+5z^{-2}+11z^{-3}+12z^{-4}$

$\therefore x(n) = 0, 1, 5, 11, 12\ldots$

Ans

PTO

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ROC $|z| < |a|$

$$1+4z-3z^2+z^3 \;\big)\; 2z-7z^2+34z^3-159z^4 \;\big(\; 2z+z^2$$

$$-2z+8z^2-6z^3+2z^4$$
(−) (−) (+) (−)

$$-7z^2+6z^3-2z^4$$
$$7z^2-28z^3+21z^4-7z^5$$
(+) (+) (−) (+)

$$34z^3-23z^4+7z^5$$
$$+34z^3+136z^4-102z^5+34z^6$$
(−) (−) (+) (−)

$$-159z^4+109z^5-34z^6$$
$$-159z^4-636z^5+477z^6-159z^7$$
(+) (+) (−) (+)

$$745z^5-511z^6+159z^7$$

$\therefore x(z) = 0+2z-7z^2+34z^3-159z^4+\ldots$

(ii) Partial Fraction Method

Q) Determine the inverse z-transform $x(z) = \dfrac{1}{1+\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}$

when, ROC $|z|>1$, and ROC $|z|<\frac{1}{2}$

Sol^n

$$x(z) = \frac{1}{1+\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}$$
$$= \frac{1}{z^{-2}\left(z^2+\frac{3}{2}z+\frac{1}{2}\right)}$$

$$\text{Or, } x(z) = \frac{z^2}{z^2+\frac{3}{2}z+\frac{1}{2}}$$

$$\text{Or, } \frac{x(z)}{z} = \frac{z}{(z+1)(z+\frac{1}{2})}$$

Let, $\dfrac{x(z)}{z} = \dfrac{z}{(z+1)(z+\frac{1}{2})} = \dfrac{A}{(z+1)} + \dfrac{B}{(z+\frac{1}{2})}$

$$A = (z+1)\times\frac{x(z)}{z}\Big|_{z=-1}$$
$$= (z+1)\times\frac{z}{(z+1)(z+\frac{1}{2})} = \frac{-1}{\frac{1}{2}-1} = +2$$

$$B = (z+\tfrac{1}{2})\times\frac{x(z)}{z}\Big|_{z=-\frac{1}{2}}$$
$$= (z+\tfrac{1}{2})\times\frac{z}{(z+1)(z+\frac{1}{2})}$$
$$= \frac{-\frac{1}{2}}{-\frac{1}{2}+1} = \frac{-\frac{1}{2}}{\frac{1}{2}} = -1$$

Now, $\dfrac{x(z)}{z} = \dfrac{+2}{(z+1)} + \dfrac{-1}{z+\frac{1}{2}}$

$$\text{Or, } x(z) = +2\cdot\frac{z}{z+1} + (-1)\times\frac{z}{z+\frac{1}{2}}$$
$$= \frac{2z}{z-(-1)} - \frac{z}{z-(-\frac{1}{2})}$$

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१ भन्दा ठूलो → denominator always ठूलो.

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For ROC $|z| > 1$,

Taking inverse z-transform,

$$x(n) = 2(-1)^n u(n) - \left(\frac{-1}{2}\right)^n u(n)$$

For ROC $|z| < 1/2$,

$$x(n) = 2(-1)^n \cdot u(-n-1) + \left(\frac{-1}{2}\right)^n \cdot u(-n-1)$$

For ROC $-1 < |z| < -1/2$;

The signal $\frac{2z}{z+1}$ will be causal and $\frac{z}{z+1/2}$ will be anti-causal.

$$x(n) = 2(-1)^n u(n) + \left(\frac{-1}{2}\right)^n \cdot u(-n-1)$$

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$n u(n) = \frac{z}{(z-1)^2}$

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Q) Determine the causal signal $x(n)$ having z-transform

$$x(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2}$$

Soln

$$x(z) = \frac{1}{(1-2z^{-1})\left(1-\frac{1}{z}\right)^2}$$

$$= \frac{z^2}{(1-2z^{-1})(z-1)^2}$$

$$= \frac{z^3}{(z-2)(z-1)^2}$$

$$\frac{x(z)}{z} = \frac{z^2}{(z-2)(z-1)^2}$$

$$\text{let, } \frac{x(z)}{z} = \frac{A}{z-2} + \frac{B}{(z-1)} + \frac{C}{(z-1)^2}$$

For A, $A = (z-2) \times \frac{x(z)}{z}\Big|_{z=2}$

$$= (z-2) \times \frac{z^2}{(z-2)(z-1)^2}$$

$$= \frac{4}{1^2} \quad \therefore A = 4.$$

For B, $B = \frac{1}{(l-i)!} \frac{d^{l-i}}{dz^{l-i}} \left[(z-p)^l \times \frac{x(z)}{z}\right]_{z=p}$

$l \rightarrow$ total degree
$i \rightarrow$ which term? no. of term

$$= \frac{1}{1!} \frac{d}{dz} \frac{z^2}{(z-2)(z-1)^2}\Big|_{z=1}$$

$$= \frac{(z-2) \times 2z - z^2}{(z-2)^2}\Big|_{z=1}$$

$$= \frac{(-1) \times 2 - 1}{(-1)^2} = -3$$

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$u(n) \leftrightarrow \frac{z}{z-1}$

$nu(n) \leftrightarrow -z \times \frac{d}{dz}\left(\frac{z}{z-1}\right)$ Date Page

$= -z \times \frac{(z-1) - z}{(z-1)^2} = \frac{z}{(z-1)^2}$

For C, $C = -1$

So, $\frac{X(z)}{z} = \frac{4}{(z-2)} + \frac{-3}{z-1} + \frac{-1}{(z-1)^2}$

$\therefore X(z) = \frac{4z}{z-2} + \frac{-3z}{z-1} - \frac{z}{(z-1)^2}$

Inverse z-transform:

$x(n) = 4(-2)^n u(n) - 3u(n) - nu(n)$

Q) Determine the causal signal if its z-transform $x(z)$ is given by,

$x(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$

$= \frac{1+\frac{3}{z}}{1+\frac{3}{z}+\frac{2}{z^2}}$

$= \frac{z+3/z}{z^2+3z+2/z^2}$

Or, $\frac{x(z)}{z} = \frac{z(z+3)}{z^2+3z+2}$

Let, $\frac{x(z)}{z} = \frac{Az+B}{z^2+3z+2}$ Or, $\frac{x(z)}{z} = \frac{(z+3)}{(z+1)(z+2)}$

Let, $\frac{x(z)}{z} = \frac{A}{z+1} + \frac{B}{z+2}$

$A = (z+1) \times \frac{x(z)}{z}\Big|_{z=-1}$

$= (z+1) \times \frac{(z+3)}{(z+1)(z+2)}\Big|_{z=-1}$

$= \frac{-1+3}{-1+2} = 2$

$B = (z+2) \times \frac{x(z)}{z}\Big|_{z=-2}$

$= (z+2) \times \frac{(z+3)}{(z+1)(z+2)}\Big|_{z=-2}$

$= \frac{3-2}{-2+1} = -1$

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So, $\frac{x(z)}{z} = \frac{2}{z+1} - \frac{1}{z+2}$

Or, $x(z) = \frac{2z}{z+1} - \frac{z}{z+2}$

$= 2 \times \frac{z}{z-(-1)} - \frac{z}{z-(-2)}$

$\therefore x(z) = 2(-1)^n u(n) - (-2)^n u(n)$ Ans

Q) Find inverse z-transform of $x(z) = \log(1+az^{-1})$.

soln

$x(n) \xleftrightarrow{ZT} x(z)$

$nx(n) \xleftrightarrow{ZT} -z \frac{d}{dz} x(z)$

For, $x(z) = \log(1+az^{-1})$

So, $nx(n) \xleftrightarrow{ZT} -z \frac{d}{dz} \log(1+az^{-1})$

$= -z \times \frac{1}{1+az^{-1}} \times a \times (-1) z^{-2}$

$nx(n) \xleftrightarrow{ZT} \frac{az^{-1}}{1+az^{-1}}$ — (i)

Now,

$u(n) \xleftrightarrow{ZT} \frac{z}{z-1}$

$a^n u(n) \xleftrightarrow{} \frac{z}{z-a}$

$a^n u(n) \xleftrightarrow{} \frac{1}{1-az^{-1}}$

$\frac{z}{z-(-a)} = (-1)^n a^n u(n)$.

$(-1)^n a^n u(n) \xleftrightarrow{} \frac{1}{1+az^{-1}}$

~~$(-1)^n a^n u(n)$~~

$(-1)^{n-1} a^{n-1} u(n-1) \xleftrightarrow{} \frac{z^{-1}}{1+az^{-1}}$

$a(-1)^{n-1} a^{n-1} u(n-1) \xleftrightarrow{ZT} \frac{az^{-1}}{1+az^{-1}}$ — (ii)

From (i) & (ii), we can write,

$nx(n) = a \cdot a^{n-1} (-1)^{n-1} u(n-1)$

$x(n) = -\frac{a^n (-1)^n u(n-1)}{n}$ Ans

Q) Find all possible inverse z-transform of $x(z) = \frac{z}{(z+1)^2 (z+2)^3}$

soln

$\frac{x(z)}{z} = \frac{1}{(z+1)^2 (z+2)^3}$

Let, $\frac{x(z)}{z} = \frac{A}{(z+1)} + \frac{B}{(z+1)^2} + \frac{C}{(z+2)} + \frac{D}{(z+2)^2} + \frac{E}{(z+2)^3}$

Now, $B = \frac{x(z)}{z} \times (z+1)^2 \Big|_{z=-1}$

$= \frac{1}{(z+1)^2 (z+2)^3} \times (z+1)^2 \Big|_{z=-1}$

$\therefore B = 1$

$E = \frac{x(z)}{z} \times (z+2)^3 \Big|_{z=-2}$

$= \frac{1}{(z+1)^2 \times (z+2)^3} \times (z+2)^2 \Big|_{z=-2}$

$\therefore E = 1$

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$$C = \frac{1}{(l-i)!}\frac{d^{l-i}}{dz}(z-p)^l \times \frac{x(z)}{z}$$

l → no. of degree/ length of pole
i → no of term

$$A = \frac{1}{(2-1)!}\frac{d^{2-1}}{dz}(z+1)^2 \times \frac{x(z)}{z}\Big|_{z=-1}$$

$$= \frac{1}{1!}\frac{d}{dz}\left[(z+1)^2 \times \frac{1}{(z+1)^2(z+2)^3}\right]\Big|_{z=-1}$$

$$= \frac{d}{dz}\left[\frac{1}{(z+2)^3}\right]\Big|_{z=-1} \qquad = (z+2)^{-3}$$

$$= \frac{(z+2)^3 - 3(z+2)^2(z+1)}{(z+2)^6}\Big|_{z=-1} \qquad = -3(z+2)^{-4}\Big|_{-1}$$

$$= \frac{(z+2)^3 - 3(z+2)^2(z+1)}{(z+2)^6}\Big|_{z=-1} \qquad = -3(-1+2)^{-4}$$

$$= -1 - (1\times 1) \qquad = -3$$

$$A = -2$$

$\frac{z}{z-a}$ → causal, $|z| > a$
$\frac{z}{z+a}$ → Non causal. $|z| < b$

Unit-3

Frequency Response of LTI system

The n^{th} order of n discrete time LTI system difference equation can be represented as,

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{k=0}^{M} b_k x(n-k)$$

Expanding the above eqⁿ. we get,

$$a_0y(n) + a_1y(n-1) + a_2y(n-2) + \dots + a_Ny(n-N) = b_0x(n) + b_1x(n-1) + b_2x(n-2) + \dots + b_Mx(n-M)$$

Taking z-transform on both sides,

$$a_0Y(z) + a_1z^{-1}Y(z) + a_2z^{-2}Y(z) + \dots + a_Nz^{-N}Y(z) = b_0X(z) + b_1z^{-1}X(z) + b_2z^{-2}X(z) + \dots + b_Mz^{-M}X(z)$$

$$\text{Or, } Y(z)(a_0 + a_1z^{-1} + a_2z^{-2} + \dots + a_Nz^{-N}) = X(z)(b_0 + b_1z^{-1} + b_2z^{-2} + \dots + b_Mz^{-M})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1z^{-1} + b_2z^{-2} + \dots + b_Mz^{-M}}{a_0 + a_1z^{-1} + a_2z^{-2} + \dots + a_Nz^{-N}}$$

$$= \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

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Q) A causal LTI system is described by the difference eqⁿ

$$y(n) = y(n-1) + y(n-2) + x(n) + 2x(n-1)$$

Find the system function and the freq. response of the system. Plot the poles and zeros and indicate the ROC.

Sol^n

JS system Stable? → यदि unit circle ROC को range मा पर्ने भने → stable
Unit circle ROC मा पर्ने भने → unstable

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$y(n) = y(n-1) + y(n-2) + x(n) + 2x(n-1)$

Taking z-transform on both sides,

$Y(z) = z^{-1}Y(z) + z^{-2}Y(z) + X(z) + 2z^{-1}X(z)$

or, $Y(z)(1 - z^{-1} - z^{-2}) = X(z)(1 + 2z^{-1})$

or, $\frac{Y(z)}{X(z)} = \frac{(1+2z^{-1})}{(1-z^{-1}-z^{-2})}$

$= \frac{z^{-1}(z+2)}{z^{-2}(z^2-z-1)}$

$= \frac{(z+2)z}{z^2-z-1}$

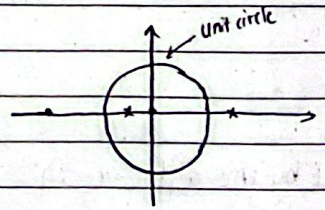
$H(z) = \frac{z(z+2)}{(z+0.618)(z-1.618)}$ $\quad \therefore H(e^{j\omega}) = \frac{e^{j\omega}(e^{j\omega}+2)}{(e^{j\omega}+0.618)(e^{j\omega}-1.618)}$

Poles are:

$z = -0.618, 1.618$

Zeros are:

$z = 0, -2$



ROC:

$|z| > 1.618$

and $|z| < 0.618$

$(z-1.618)$ causal vayera

$(z+0.618)$ non causal vayera

Q) Find the difference eqⁿ satisfying the i/p and o/p of LTI system with transfer function

$$h(z) = \frac{(1+z^{-1})^2}{(1-0.5z^{-1})(1+0.75z^{-1})}$$

Also comment on the causality of given transfer function.

Soln

$$h(z) = \frac{(1+z^{-1})^2}{(1-0.5z^{-1})(1+0.75z^{-1})}$$

or, $\frac{Y(z)}{X(z)} = \frac{[z^{-1}(z+1)]^2}{z^{-1}(z+0.5) \times z^{-1}(z+0.75)}$

$= \frac{(z+1)^2}{(z+0.5)(z+0.75)}$

or, $\frac{Y(z)}{X(z)} = \frac{(1+z^{-1})^2}{(1-0.5z^{-1})(1+0.75z^{-1})}$

or, $Y(z)(1+0.75z^{-1}-0.5z^{-1}-0.375z^{-2}) = X(z)(1+2z^{-1}+z^{-2})$

or, $Y(z) + 0.75z^{-1}Y(z) - 0.5z^{-1}Y(z) - 0.375z^{-2}Y(z) = X(z) + 2z^{-1}X(z) + z^{-2}X(z)$

Taking Inverse z-transform,

$y(n) + 0.75y(n-1) - 0.5y(n-1) - 0.375y(n-2) = x(n) + 2x(n-1) + x(n-2)$

$\therefore y(n) = -0.25y(n-1) + 0.375y(n-2) + x(n) + 2x(n-1) + x(n-2)$

This is the reqd difference eqⁿ.

As in the difference eqⁿ, the system's output depend on present and past values only. so, the system is causal. Ans

Q) An LTI system is described by the difference eqⁿ

$y(n) - \frac{9}{4}y(n-1) + \frac{1}{2}y(n-2) = x(n) - 3x(n-1)$

Specify the ROC of $h(z)$ and determine $h(n)$ for the following condition

(i) the system is stable (ii) the system is causal

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Soln

$$y(n) - \frac{9}{4}y(n-1) + \frac{1}{2}y(n-2) = x(n) - 3x(n-1)$$

Taking z-transform on both sides,

$$Y(z) - \frac{9}{4}z^{-1}Y(z) + \frac{1}{2}z^{-2}Y(z) = X(z) - 3z^{-1}X(z)$$

$$\text{or, } Y(z)\left(1 - 2.25z^{-1} + 0.5z^{-2}\right) = X(z)\left(1 - 3z^{-1}\right)$$

$$\text{or, } \frac{Y(z)}{X(z)} = \frac{(1-3z^{-1})}{(1-2.25z^{-1}+0.5z^{-2})}$$

$$= \frac{z^{-1}(z-3)}{z^{-2}(-2.25z+0.5)+z^{2})}$$

$$\text{or, } H(z) = \frac{z(z-3)}{(-2.25z+0.5)(z^2-2.25z+0.5)}$$

$$\text{or, } H(z) = \frac{z(z-3)}{(z-0.25)(z-2)}$$

$$\text{or, } \frac{H(z)}{z} = \frac{(z-3)}{(z-0.25)(z-2)}$$

Let, $\frac{H(z)}{z} = \frac{A}{(z-0.25)} + \frac{B}{(z-2)}$

$$A = \frac{H(z)}{z} \times (z-0.25)\Big|_{z=0.25}$$

$$= \frac{(z-3)}{(z-0.25)(z-2)} \times (z-0.25)\Big|_{z=0.25}$$

$\therefore A = 1.57$

Also, $B = \frac{H(z)}{z} \times (z-2)\Big|_{z=2}$

$$= \frac{(z-3)(z-2)}{(z-0.25)(z-2)}\Big|_{z=2}$$

$\therefore B = -0.57$

$u(n) \rightarrow$ causal | $u(-n-1)$

$$\frac{H(z)}{z} = \frac{1.57}{(z-0.25)} - \frac{0.57}{(z-2)}$$

$$\therefore H(z) = \frac{1.57z}{(z-0.25)} - \frac{0.57z}{(z-2)}$$

ROC : $|z| > 0.25$, $|z| > 2$

For region $0.25 < |z| < 2$,

$h(n) = 1.57(0.25)^n u(n) - 0.57 \times (2)^n u(-n-1)$ (causal & Non causal) (Non-causal)

(Stable) → unit circle.

For region $|z| > 2$,

$h(n) = 1.57(0.25)^n u(n) - 0.57(2)^n u(n)$ (causal)

Relationship of frequency response to pole zero of system

$$H(z) = k\frac{(z-z_1)(z-z_2)\cdots}{(z-p_1)(z-p_2)\cdots}$$

$$H(e^{j\omega}) = k\frac{(e^{j\omega}-z_1)(e^{j\omega}-z_2)\cdots}{(e^{j\omega}-p_1)(e^{j\omega}-p_2)}$$

$$|H(e^{j\omega})| = k\frac{\text{distance from zeros}}{\text{distance from poles}}$$

zero near unit circle → magnitude small

pole near unit circle → magnitude large

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Unit-7

Discrete Fourier Transform

The n-point DFT of a signal $x(n)$ is defined as:

$$\text{DFT}[x(n)] = X(K) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi kn}{N}} \quad \text{for } K = 0, 1, 2, \ldots N$$

$$X(K) = \sum_{n=0}^{N-1} x(n)\, W_N^{kn} \quad \text{for } K = 0, 1, 2 \ldots N-1$$

where,

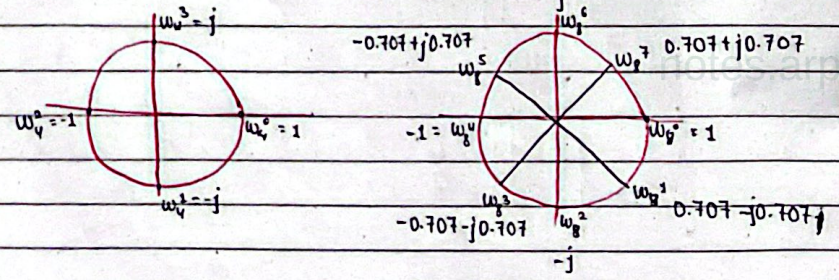
W_N = Twiddle factor = $e^{-j2\pi/N}$

Inverse Discrete Fourier Transform

The IDFT of sequence $X(K)$ is defined as:

$$\text{IDFT}[X(K)] = x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)\, e^{j2\pi kn/N} \quad \text{for } n = 0, 1, 2 \ldots N-1$$

$$x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)\, W_N^{-kn} \quad \text{for } n = 0, 1, 2 \ldots N-1$$



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$x(n) = \{1, -1, 2, -2\}$ add zero padding

Q) Find the four point DFT of the sequence $x(n) = \{1, -1, 2, -2\}$

Soln

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad \text{for } K = 0, 1, 2, 3 \qquad (N = 4)$$

For K = 0,

$$X(0) = \sum_{n=0}^{3} x(n) W_4^0$$

$$= x(0)W_4^0 + x(1)W_4^0 + x(2)W_4^0 + x(3)W_4^0$$
$$= 1 \times 1 + (-1) \times 1 + 2 \times 1 + (-2) \times 1$$
$$= 0$$

For K = 1,

$$X(1) = \sum_{n=0}^{3} x(n) W_4^n$$

$$= x(0)W_4^0 + x(1)W_4^1 + x(2)W_4^2 + x(3)W_4^3$$
$$= 1 \times (-j) + (-1) \times (-j) + 2 \times (-j) + (-2) \times (-j)$$
$$= 1 \times 1 + (-1) \times (-j) + 2 \times (-1) + (-2) \times j$$
$$= -1 - 1j$$

For K = 2,

$$X(2) = \sum_{n=0}^{3} x(n) W_4^{2n}$$

$$= x(0)W_4^0 + x(1)W_4^2 + x(2)(W_4^2)^2 + x(3)(W_4^3)^2$$
$$= 1 \times 1 + (-1) \times (-1) + 2 \times 1 + (-2) \times j^2$$
$$= 1 + 1 + 2 + (-2) \times (-1)$$
$$= 6$$

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For K = 3,

$$X(3) = \sum_{n=0}^{3} x(n)(W_4^n)^3$$

$$= x(0)W_4^0 + x(1)(W_4^1)^3 + x(2)(W_4^2)^3 + x(3)(W_4^3)^3$$

$$= 1\times 1 + (-1)\times(-j)^3 + 2\times(-1)^3 + (-2)\times(j)^3$$

$$= 1 + (-1)j - 2 + 2j$$

$$= -1 + j$$

$$X(k) = [0, -1-j, 6, -1+j]$$ Ans

DFT using matrix method

$$X(k) = [W_N^{kn}][x(n)]$$

Q) N = 4

$$X(K) = [W_N^{Kn}]_{4\times4}[x(n)]_{4\times1}$$

$$X(k) = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} x(n) \\ \\ \\ \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1+2-2 \\ 1+j-2-2j \\ 1+1+2+2 \\ 1-j-2+2j \end{bmatrix} = \begin{bmatrix} 0 \\ -1-j \\ 6 \\ -1+j \end{bmatrix}$$ Ans

Q) Find four point DFT of sequence $x(n) = \{1, 2, 0, 1\}$.

Soln

N = 4

$$X(K) = [W_N^{Kn}]_{4\times4}[x(n)]_{4\times1}$$

$$= \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+2+0+1 \\ 1-2j+j \\ 1-2-1 \\ 1+2j-j \end{bmatrix} = \begin{bmatrix} 4 \\ 1-j \\ -2 \\ 1+j \end{bmatrix}$$ Ans

Properties of DFT

(i) Periodicity

If DFT $[x(n)] = X(k)$

then,

$x(n-N) = x(n)$

$X(k+N) = X(k)$

Proof:

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By definition,

$$DFT[x(n)] = X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}}$$

$$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-\frac{-j2\pi(K+N)n}{N}}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}} \cdot e^{-j\frac{2\pi KN}{N}}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j}$$

$$= X(K)$$

(ii) Linearity

If $DFT[x_1(n)] = X_1(K)$
and $DFT[x_2(n)] = X_2(K)$
then, $DFT[ax_1(n) + bx_2(n)] = aX_1(K) + bX_2(K)$

Proof:

$$DFT[ax_1(n) + bx_2(n)] = \sum_{n=0}^{N-1} [ax_1(n) + bx_2(n)]\, e^{-j2\pi Kn/N}$$

$$= a\sum_{n=0}^{N-1} x_1(n) e^{-\frac{j2\pi Kn}{N}} + b\sum_{n=0}^{N-1} x_2(n) e^{-\frac{j2\pi Kn}{N}}$$

$$= aX_1(K) + bX_2(K)$$

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(iii) Time Reversal Property

If $DFT[x(n)] = X(K)$
then, $DFT[x(-n), \bmod N] = DFT[x(N-n)]$
$= X[(-K), \bmod N] = X(N-K)$

Proof:

$$DFT[x(n)] = X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi Kn/N}$$

$$DFT[x(N-n)] = \sum_{n=0}^{N-1} x(N-n)\, e^{-j2\pi Kn/N}$$

Let, $m = N-n$, then $n = N-m$

$$DF[x(N-n)] = \sum_{m=0}^{N-1} x(m)\, e^{-j2\pi K(N-m)/N}$$

$$= \sum_{m=0}^{N-1} x(m)\, e^{-\frac{j2\pi KN}{N}} \cdot e^{\frac{j2\pi Km}{N}}$$

$$= \sum_{m=0}^{N-1} x(m)\, e^{\frac{j2\pi Km}{N}}$$

$$= \sum_{m=0}^{N-1} x(m)\, e^{\frac{j2\pi Km}{N}} \cdot e^{-j2\pi m}$$

$$= \sum_{m=0}^{N-1} x(m)\, e^{-j2\pi m\left(\frac{N-K}{N}\right)}$$

$$= X(N-K)$$

(iv) Circular Frequency Shift

If $DFT[x(n)] = X(K)$
then,

$$DFT[x(n)\, e^{\frac{j2\pi ln}{N}}] = X[(K-l), \bmod N]$$

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Proof:

$$\text{DFT}[x(n)] = X(K) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi Kn}{N}}$$

$$\text{DFT}\left[x(n)\, e^{\frac{j2\pi ln}{N}}\right] = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi Kn}{N}} \cdot e^{\frac{2\pi ln}{N}}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi n(K-l)}{N}}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi n(N+K-l)}{N}}$$

$$= X(N+K-l)$$

$$= X[(K-l), \text{mod } N]$$

(v) Circular Convolution of two sequence

The convolution property of DFT says that multiplication of DFT's of two sequences is equivalent to the DFT of the circular convolution of the two sequences.

If $\text{DFT}[x_1(n)] = X_1(K)$
and $\text{DFT}[x_2(n)] = X_2(K)$
then,

$$X_1(K)\, X_2(K) = \text{DFT}[x_1(n) \oplus x_2(n)]$$

Proof:

Let $x_1(n)$ and $x_2(n)$ be two finite sequences of length n. Then the n-point DFT is given by

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n)\, e^{\frac{-j2\pi Kn}{N}} \quad \text{for } K = 0, 1, \ldots N-1 \quad \text{——(i)}$$

$$X_2(K) = \sum_{l=0}^{N-1} x_2(l)\, e^{\frac{-j2\pi Kl}{N}} \quad \text{for } K = 0, 1, \ldots N-1 \quad \text{——(ii)}$$

On multiplying the above two DFT's, another DFT is obtained say $X_3(K)$. Therefore, $X_3(K)$ will be N-point DFT of sequence $x_3(n)$.

$$\therefore X_3(K) = X_1(K) \times X_2(K) \quad \text{——(iii)}$$

Now, By using inverse Discrete Fourier Transform,

$$x_3(m) = \frac{1}{N} \sum_{K=0}^{N-1} X_3(K)\, e^{\frac{j2\pi Km}{N}}$$

$$= \frac{1}{N} \sum_{K=0}^{N-1} X_1(K)\, X_2(K)\, e^{\frac{j2\pi Km}{N}} \quad \text{——(iv)}$$

Substituting the value of (i) & (ii) in eqⁿ (iv), we get,

$$x_3(m) = \frac{1}{N} \sum_{K=0}^{N-1} \sum_{n=0}^{N-1} x_1(n)\, e^{\frac{-j2\pi Kn}{N}} \sum_{l=0}^{N-1} x_2(l)\, e^{\frac{-j2\pi Kl}{N}} \cdot e^{\frac{j2\pi Km}{N}}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \sum_{K=0}^{N-1} e^{\frac{j2\pi K(m-n-l)}{N}} \quad \text{——(v)}$$

Let, $m-n-l = Np$ where p is integer.

Then,

$$e^{\frac{j2\pi K(m-n-l)}{N}} = e^{\frac{j2\pi KNp}{N}} = e^{j2\pi Kp} = (e^{j2\pi p})^K = (1)^K$$

$$\sum_{K=0}^{N-1} e^{\frac{j2\pi (m-n-l)}{N}} = \sum_{K=0}^{N-1} (1)^K = N$$

Now, From eqⁿ (v)

$$x_3(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \times N$$

$$= \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \quad \text{——(vi)}$$

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If $x_2(l)$ is a periodic sequence with periodicity of N-samples.

$$x_2(l \pm pN) = x_2(l)$$

Here, $m - n - l = Np$

or, $l = m - n - Np$

Then,

$$x_2(l) = x(m-n-Np) = x(m-n)$$
$$= x[(m-n), \text{mod } N]$$

Then, from eqn (vi),

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) \sum_{n=0}^{N-1} x_2(m-n)$$
$$= \sum_{n=0}^{N-1} x_1(n)\, x_2(m-n)$$

Replacing m by n and n by K.

$$x_3(n) = \sum_{K=0}^{N-1} x_1(K)\, x_2(n-K)$$

Hence, we can conclude that multiplication of DFTs of two sequence is equivalent to the DFT of circular convolution of two sequences.

Fast Fourier Transform (FFT)

FFT

- DIT FFT (Decimation in Time) (Twiddle factor $e^{\pm j2\pi/N}$)
- DIF FFT (Decimation in Frequency) (Twiddle factor $e^{\pm j2\pi/N}$)

(0) 00 → 00 0
(1) 01 → 10 2
(2) 10 → 01 1
(3) 11 → 11 3

0 000 → 000 0
1 001 → 100 4
2 010 → 010 2
3 011 → 110 6
4 100 → 001 1
5 101 → 101 5
6 110 → 011 3
7 111 → 111 7

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Butterfly Diagram Method

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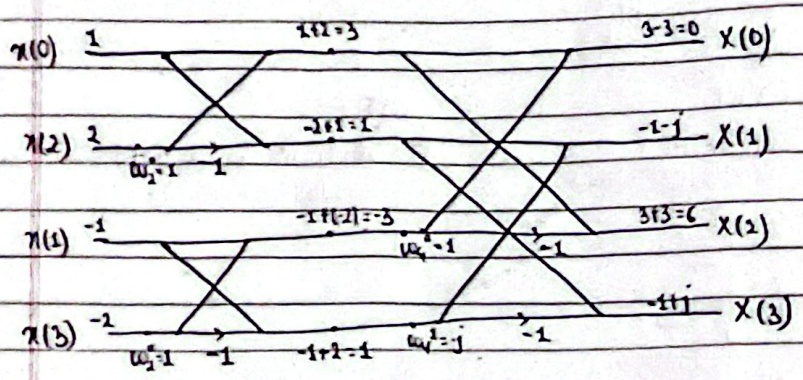
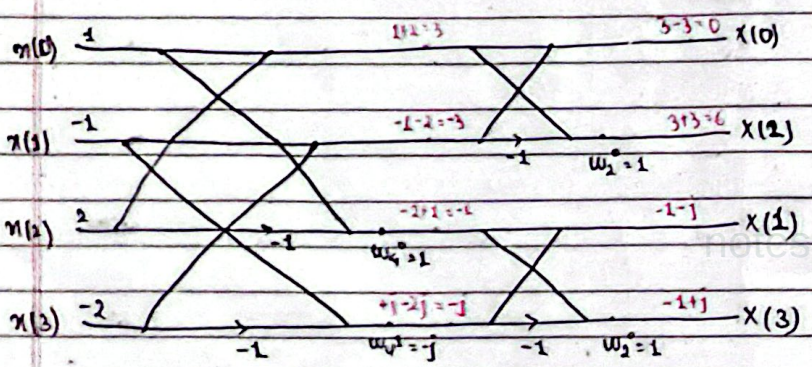
Q) Find DIT-FFT of $x(n) = \{1, -1, 2, -2\}$.

Soln



Q) Find DIF-FFT of $x(n) = \{1, -1, 2, -2\}$



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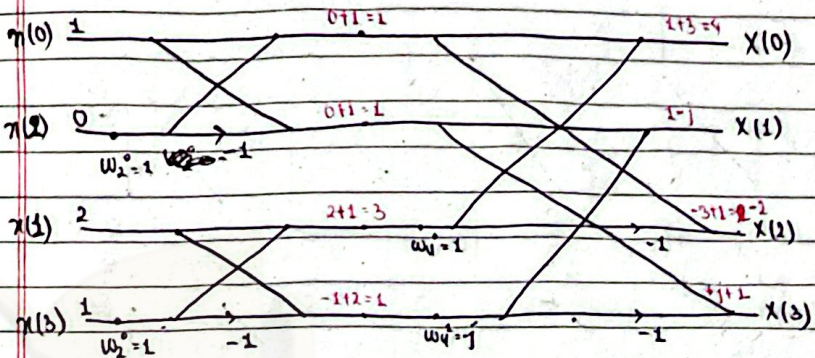
Q) Find DIT-FFT of $x(n) = \{1, 2, 0, 1\}$



Q) Find the 8 point DIT FFT of $x(n) = \sin \frac{n\pi}{3}$.

Here,

$$x(n) = \sin \frac{n\pi}{3}$$

For 8-point, (0 to 7) → n.

$x(n) = \{0, 0.86, 0.86, 0, -0.86, -0.86, 0, 0.86\}$

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DIF FFT :

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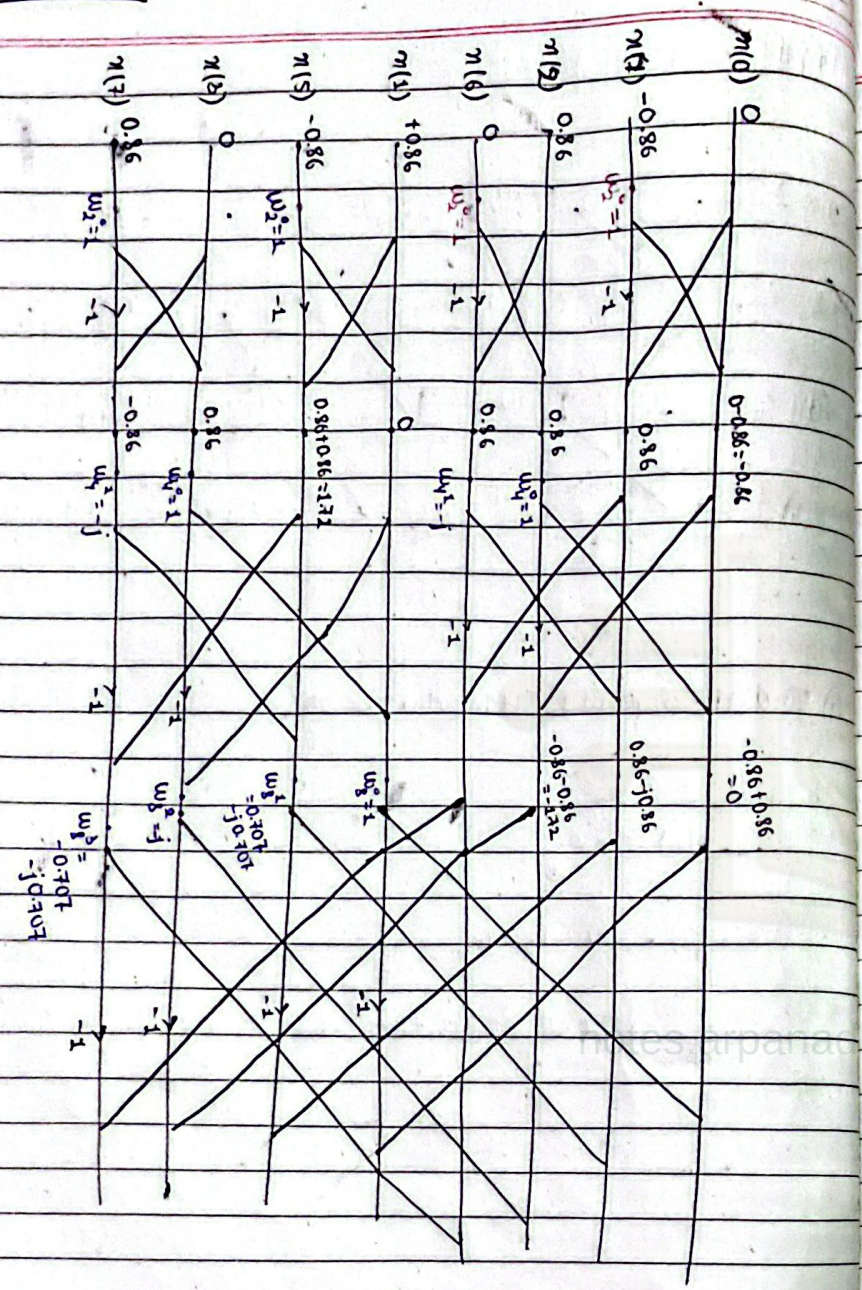

DIF FFT :

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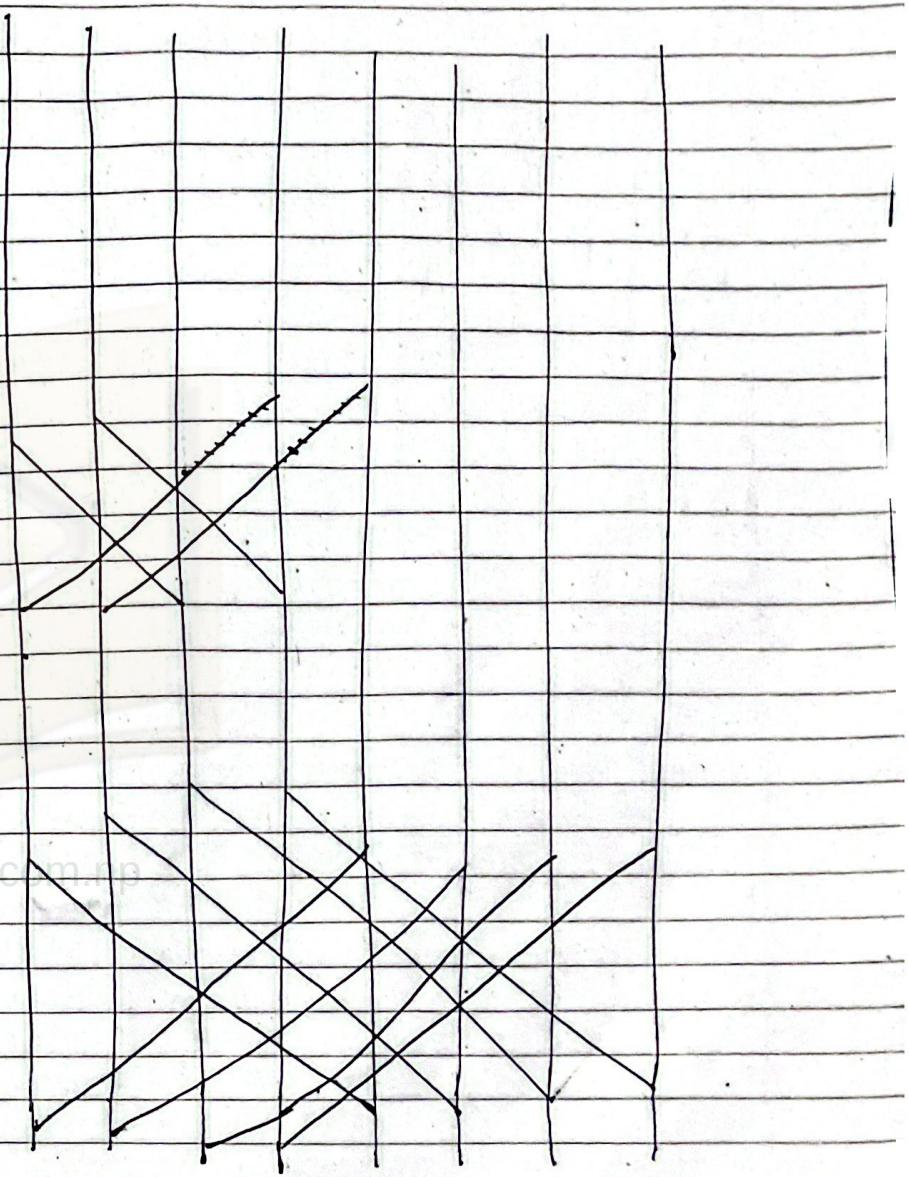

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Unit-4

FILTER STRUCTURE

(Infinite Impulse Response) IIR Filter	(Finite Impulse Response) FIR Filter
$H(z) = \frac{\sum_{k=0}^{N} b_k z^{-k}}{1 + \sum_{k=1}^{M} a_k z^{-k}}$	$H(z) = \sum_{k=0}^{N} b_k z^{-N}$
- only poles	- only zeros
- both poles and zeros	
(i) Direct form I	(i) Direct form
(ii) Direct form II	(ii) Cascade
(iii) Cascade	(iii) Lattice
(iv) Parallel	
(v) Lattice	
(vi) Lattice ladder	

Q) Obtain direct form-I and direct form-II of the system described by
$y(n) = -0.1\,y(n-1) + 0.72\,y(n-2) + 0.7\,x(n) - 0.252\,x(n-2)$

Soln

$y(n) = -0.1\,y(n-1) + 0.72\,y(n-2) + 0.7\,x(n) - 0.252\,x(n-2)$

Taking z-transform on both sides,

$Y(z) = -0.1\,z^{-1}Y(z) + 0.72\,z^{-2}Y(z) + 0.7\,X(z) - 0.252\,z^{-2}X(z)$

$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{0.7 - 0.252\,z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$

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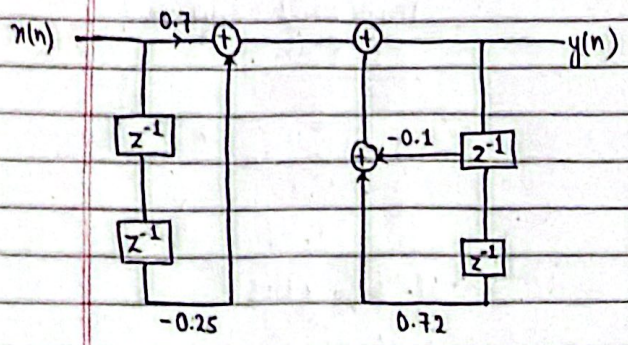
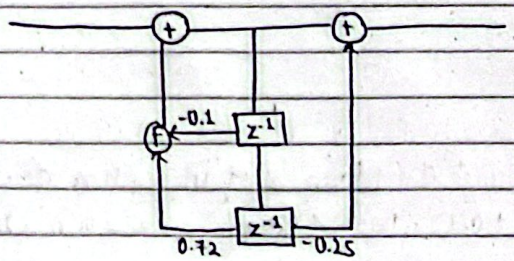
Direct Form I



Direct form II:



a) Obtain direct form I and II and cascade realization for the following system $y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-1) + 0.6x(n-2)$

3rd

$y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-1) + 0.6x(n-2)$

Taking z-transform on both sides,

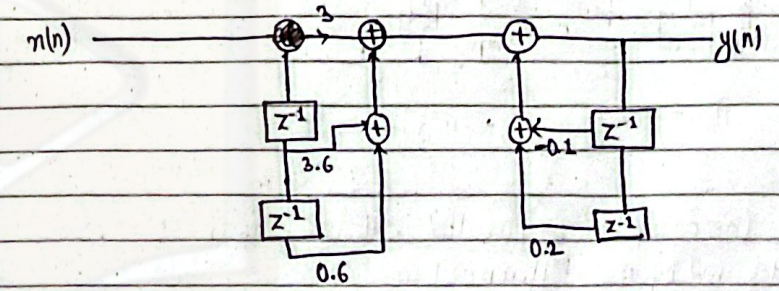
$Y(z) = -0.1z^{-1}Y(z) + 0.2z^{-2}Y(z) + 3X(z) + 3.6z^{-1}X(z) + 0.6z^{-2}X(z)$

or, $H(z) = \frac{Y(z)}{X(z)} = \frac{1+0.1z^{-1}-0.2z^{-2}}{}$

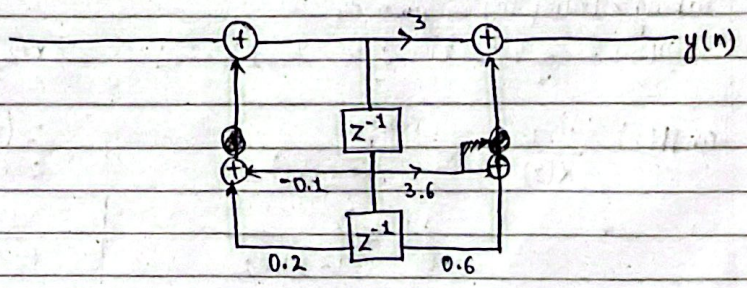
or, $Y(z) + 0.1z^{-1}Y(z) - 0.2z^{-2}Y(z) = 3X(z) + 3.6z^{-1}X(z) + 0.6z^{-2}X(z)$

or, $H(z) = \frac{Y(z)}{X(z)} = \frac{3+3.6z^{-1}+0.6z^{-2}}{1+0.1z^{-1}-0.2z^{-2}}$

Direct form I:



Direct Form II:



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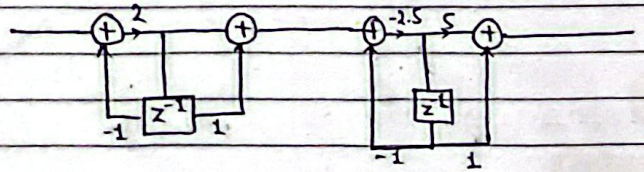
'Cascade Realization':

$$H(z) = \frac{3+3.6z^{-1}+0.6z^{-2}}{1+0.1z^{-1}-0.2z^{-2}}$$

$$= \frac{(1+z^{-1})(5+z^{-1})}{(2+z^{-1})(-2.5+z^{-1})}$$

$$= \left(\frac{1+z^{-1}}{2+z^{-1}}\right) + \left(\frac{5+z^{-1}}{-2.5+z^{-1}}\right)$$



$H_1(z)$ $H_2(z)$

Eg

Q) Determine the cascade and parallel realization of the discrete time system described by the differential eq?

$$y(n) = \frac{-3}{4}y(n-1) + \frac{1}{4}y(n-2) + x(n) + \frac{1}{2}x(n-1)$$

Soln

Taking z-transform of given eq?,

$$Y(z) = \frac{-3}{4}z^{-1}Y(z) + \frac{1}{4}z^{-2}Y(z) + X(z) + \frac{1}{2}z^{-1}X(z)$$

$$\text{Or, } H(z) = \frac{Y(z)}{X(z)} = \frac{1+\frac{1}{2}z^{-1}}{1+\frac{3}{4}z^{-1}+\frac{1}{4}z^{-2}+1} = \frac{z(z+\frac{1}{2})}{z^2+\frac{3}{4}z-\frac{1}{4}}$$

$$= \frac{1+0.5z^{-1}}{(z^{-1}-4)(z^{-1}+1)} = \frac{z(z+\frac{1}{2})}{(z+1)(z-\frac{1}{4})}$$

$$\text{Let, } \frac{H(z)}{z} = \frac{A}{(z+1)} + \frac{B}{(z-\frac{1}{4})}$$

$$\text{Here, } A = (z+1)\times \frac{(z+\frac{1}{2})}{(z+1)(z-\frac{1}{4})}\Big|_{z=-1}$$

$$= \frac{-0.5}{-1.25}$$

$$= 2/5$$

$$\text{Also, } B = \left(z-\frac{1}{4}\right)\times \frac{(z+\frac{1}{2})}{(z+1)(z-\frac{1}{4})}\Big|_{z=\frac{1}{4}}$$

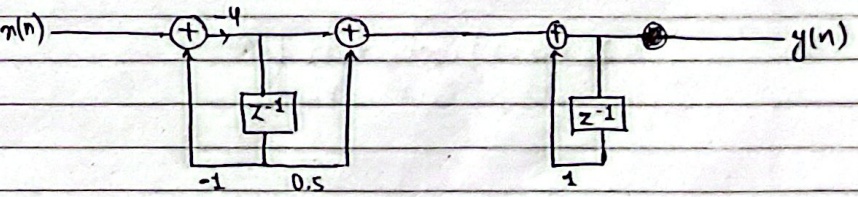
$$= \frac{0.75}{1.25}$$

$$= 3/5$$

$$\text{Hence, } H(z) = \frac{2/5\,z}{(z+1)} + \frac{3/5\,z}{z-\frac{1}{4}}$$

$$= \frac{2/5}{1+z^{-1}} + \frac{3/5}{1-\frac{1}{4}z^{-1}}$$

For cascade, $H(z) = \underbrace{\frac{1+0.5z^{-1}}{z^{-1}-4}}_{H_1(z)} \times \underbrace{\frac{1}{z^{-1}+1}}_{H_2(z)}$



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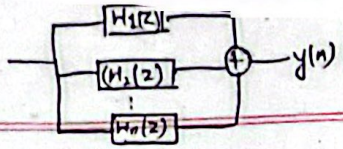
(Parallel R in course)



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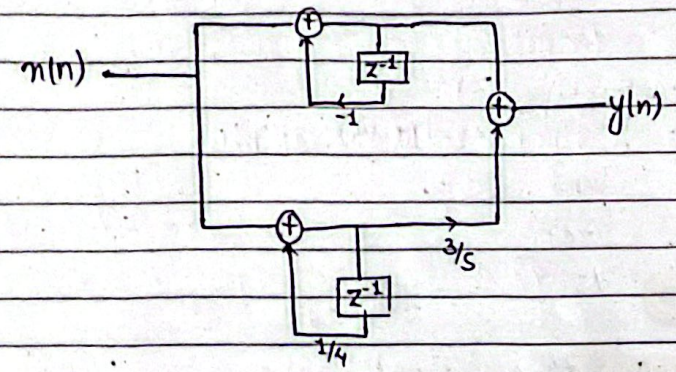
For parallel,

$$H(z) = \frac{2/5}{1+z^{-1}} + \frac{3/5}{1-1/4 z^{-1}}$$



Q) Find parallel realization

$$H(z) = \frac{3(2z^2+5z+4)}{(2z+1)(z+2)}$$

Sol^n

$$H(z) = \frac{6z^2+15z+12}{2z^2+5z+2}$$

$$2z^2+5z+2 \;\big)\; 6z^2+15z+12 \;\big(\; 3$$
$$\underline{6z^2+15z+6}$$
$$6$$

$$H(z) = \frac{6}{2z^2+5z+2} + 3$$

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$$H(z) = 3 + \frac{6}{(2z+1)(z+2)}$$

$$= 3 + H_1(z)$$

where, $H_1(z) = \frac{6}{(2z+1)(z+2)}$

Let, $H_1(z) = \frac{A}{(2z+1)} + \frac{B}{(z+2)}$

Now, $A = (2z+1) \times \frac{6}{(2z+1)(z+2)}\Big|_{z=-1/2}$

$$= \frac{6}{2-1/2}$$

$$\therefore A = 4$$

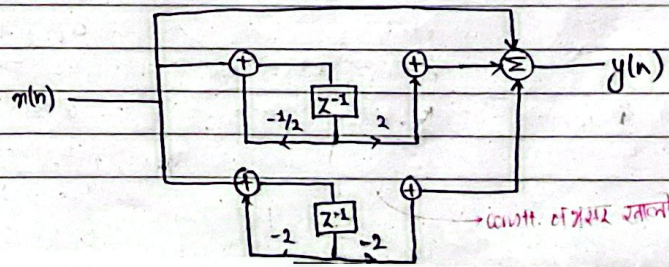
Also, $B = (z+2) \times \frac{6}{(2z+1)(z+2)}\Big|_{z=-2}$

$$= \frac{6}{-4+1}$$

$$= -2$$

Then, $H(z) = 3 + \frac{4}{2z+1} + \frac{-2}{z+2}$

$$= 3 + \frac{2z^{-1}}{1+1/2 z^{-1}} + \frac{-2z^{-1}}{1+2z^{-1}}$$



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IIR Filters v/s FIR Filter Direct form I v/s II

Quantization effect => short note

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Lattice v/s Lattice ladder,

Conversion from lattice structure to direct form structure

$a_m(0) = 1$

$a_m(m) = k_m$

$a_m(k) = a_{m-1}(k) + a_m(m)\, a_{m-1}(m-k)$

Conversion from direct form to lattice structure

$a_m(0) = 1$

$a_m(m) = k_m$

$$a_{m-1}(k) = \frac{a_m(k) - a_m(m)\, a_m(m-k)}{1 - a_m^2(m)}$$

For ladder,

$$c_m = b_m - \sum_{i=m+1}^{M} c_i a_i(i-m) \quad , m = M, M-1, M-2 \ldots 0$$

2025 spring (old)

Q4(a) Draw the lattice ladder structure for the following structure:

$$H(z) = \frac{1 - 0.8z^{-1} - 0.15z^{-2}}{1 + 0.1z^{-1} - 0.7z^{-2}}$$

$\rightarrow (1 + a_1 z^{-1} + a_2 z^{-2})$

$$\because H(z) = \frac{\sum_{k=0}^{N} b_k z^{-k}}{1 + \sum_{k=1}^{M} a_k z^{-k}} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

$a_1 = 0.1$
$a_2 = -0.7$

$a_1(1) = 0.1$
$a_2(2) = -0.7$

M → Numerator order
m → denominator order

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$m = 2$

$a_2(1) = 0.1$

$a_2(2) = -0.7 = k_2$

For, $a_1(1)$, $m = 2, k = 1$

$$a_1(1) = \frac{a_2(1) - a_2(2)\, a_2(1)}{1 - a_2^2(2)}$$

$$= \frac{0.1 - [(-0.7) \times 0.1]}{1 - (-0.7)^2}$$

$$= 0.33$$

If lattice coefficient k_1, k_2 are less than 1, system is STABLE!

Again, $b_0 = 1$

$b_1 = -0.8$

$b_2 = -0.15$

$M = 2$

$m = 2, 1, 0$

For $m = 2$,

$$c_2 = b_2 - \sum_{i=2+1}^{2} c_i a_i(i-m)$$

$$= b_2 + 0$$

$$= -0.15$$

For $m = 1$,

$$c_1 = b_1 - \sum_{i=1+1}^{2} c_i a_i(i-m)$$

$$= -0.8 - [c_2 a_2(2-1)]$$

$$= -0.8 - c_2 a_2(1)$$

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$= -0.8 - (-0.15 \times 0.1)$

$= -0.785$

For m=0,

$$c_0 = b_0 - \sum_{i=0+1}^{2} c_i a_i (i-m)$$

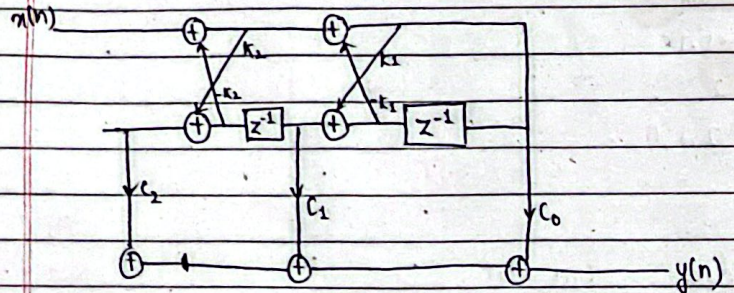
$= 1 - [c_1 a_1 (1-0) + c_2 a_2 (2-0)]$

$= 1 - [c_1 a_1 (1) + c_2 a_2 (2)]$

$= 1 - [-0.785 \times 0.33 + (-0.15 \times -0.73)]$

$=$

Lattice ladder:



Q) Draw lattice ladder structure

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}}$$

Soln

Soln Here,

For lattice,

$m = 3$

$a_3(1) = 13/24$

$a_3(2) = 5/8$

$a_3(3) = 1/3 = K_3$

For K_2, $a_2(2)$, $m = 3$, $K = 2$

$$a_2(2) = \frac{a_3(2) - a_3(3) a_3(3-2)}{1 - a_3^2(3)}$$

$$= \frac{5/8 - 1/3 \times 13/24}{1 - (1/3)^2}$$

$= 1/2 = 0.5$

For K_1, $a_1(1)$, $m = 2$, $K = 1$

$$a_1(1) = \frac{a_2(1) - a_2(2) a_2(1)}{1 - a_2^2(2)}$$

Now, $a_2(1)$: $m = 3$, $K = 1$

$$a_2(1) = \frac{a_3(1) - a_3(3) \times a_3(3-1)}{1 - a_3^2(3)}$$

$$= \frac{13/24 - 1/3 \times 5/8}{1 - (1/3)^2}$$

$= 3/8 = 0.375$

Now, For K_1 : $a_1(1)$: $m = 2$, $K = 1$

$$a_1(1) = \frac{a_2(1) - a_2(2) a_2(2-1)}{1 - a_2^2(2)}$$

$$= \frac{0.375 - 0.5 \times 0.375}{1 - 0.5^2}$$

$= 1/4 = 0.25$

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For ladder,

$b_0 = 1$

$b_1 = 1$

$b_2 = 2 \quad b_3 = 1 \quad M = 3$

$m = 3, 2, 1, 0$

For m=3,

$$c_3 = b_3 - \sum_{i=3+1}^{3} c_i a_i (i-m)$$

$= 1 - 0$

$\therefore c_3 = 1$

For m=2,

$$c_2 = b_2 - \sum_{i=2+1}^{3} c_i a_i (i-m)$$

$= b_2 - [c_3 a_3 (3-2)]$

$= 2 - [1 \times 13/24]$

$\therefore c_2 = 1.458$

For m=1,

$$c_1 = b_1 - \sum_{i=1+1}^{3} c_i a_i (i-m)$$

$= b_1 - [c_2 a_2 (2-1) + c_3 a_3 (3-1)]$

$= 1 - [1.458 \times 0.375 + 1 \times 5/8]$

$= -0.17175$

$c_0 = b_0 -$

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$a_0 \to 1$.

$a_0(0) \to 1$

FIR

$$H(z) = \sum_{K=0}^{N} a_K z^{-K}$$

Q) Draw the lattice structure for the following FIR filter function.

$$H(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$$

Soln

Comparing with $H(z) = a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots$

$a_0 = 1$ $\quad a_2(0) = 1$

$a_1 = 2$ $\quad a_2(1) = 2$

$a_2 = 1/3$ $\quad a_2(2) = 1/3 = k_2$

We have, $a_{m-1}(k) = \dfrac{a_m(k) - a_m(m) \times a_m(m-k)}{1 - a_m^2(m)}$

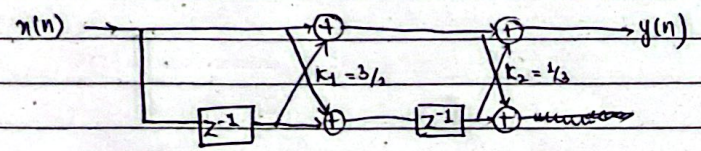
For k_1, $m = 2, k = 1$.

$$a_1(1) = \frac{a_2(1) - a_2(2) \times a_2(1)}{1 - [a_2(2)]^2}$$

$$= \frac{2 - 1/3 \times 2}{1 - 1/9}$$

$$= 3/2$$

structure:



(All multiplier coefficients are positive).

Q) Determine the lattice coefficient corresponding with the FIR system function.

$$H(z) = 1 + \frac{7}{4}z^{-1} + \frac{3}{5}z^{-2}$$

Soln

$m = 2$ $\quad$ comparing with $a_0 + a_1 z^{-1} + a_2 z^{-2}$

$\therefore a_0 = 1$ $\quad a_2(0) = 1$

$a_1 = 7/4$ $\quad a_2(1) = 7/4$

$a_2 = 3/5$ $\quad a_2(1) = 3/5$

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Q) Determine the cascade & direct structure of FIR filter having function

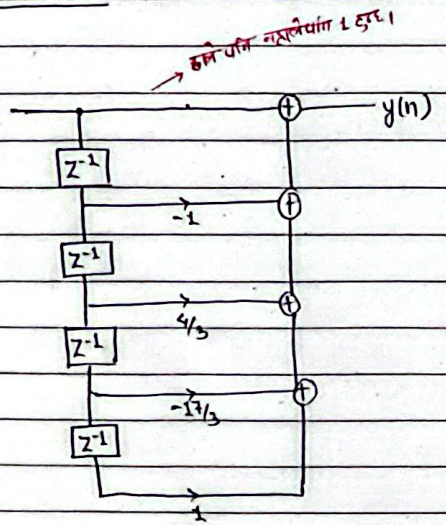
$$H(z) = (1+z^{-1}+3z^{-2})(1-2z^{-1}+\tfrac{1}{3}z^{-2})$$

Soln

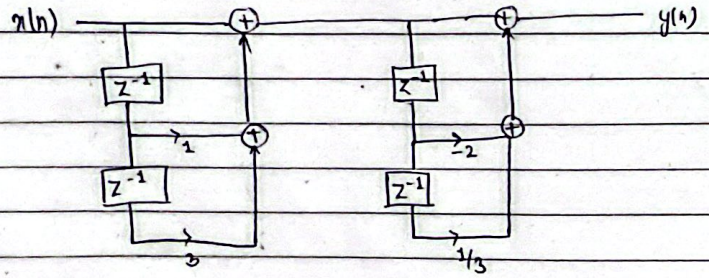
$$H(z) = 1-2z^{-1}+\tfrac{1}{3}z^{-2}-2z^{-2}+z^{-1}+\tfrac{1}{3}z^{-3}+3z^{-2}-6z^{-3}+z^{-4}$$

$$= 1-z^{-1}+\frac{4}{3}z^{-2}-\frac{17}{3}z^{-3}+z^{-4}$$

Direct structure:



Cascade structure:



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Unit-V

IIR Filter Design

i) IIM
ii) BLT

IIM (Impulse Invariance Method)

Let $h_a(t)$ = Impulse response in time-design
$h_a(nT)$ = Sample version of $h_a(t)$
$H_a(s)$ = Transfer function of analog filter
$H(z)$ = Transfer function of digital filter
Ω = Analog Frequency
ω = Digital Frequency

Now, let the system transfer function of analog filter be $H_a(s)$. $H_a(s)$ can be represented in terms of partial fraction expansion

That is, $H_a(s) = \frac{A_1}{s-P_1} + \frac{A_2}{s-P_2} + \dots$

$$H_a(s) = \sum_{i=1}^{N} \frac{A_i}{s-P_i} \quad \text{——} (1)$$

Here, A_1, A_2 are the coefficient of partial fractions and $P_1, P_2, P_3 \dots$ are the poles

Impulse response of analog filter $h_a(t)$ can be obtained from $H_a(s)$ by taking Inverse laplace transform.

$$h_a(t) = \sum_{i=1}^{N} A_i e^{P_i t} \quad \text{——} (2)$$

Now, unit impulse response for discrete structure is obtained by sampling $h_a(t)$. Thus,

$$h(n) = \sum_{i=1}^{N} A_i e^{P_i n T}$$

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The system transfer function of digital filter is obtained by taking z-transform of h(n).

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \sum_{i=1}^{N} A_i e^{p_i n T} z^{-n}$$

$$= \sum_{i=1}^{N} A_i \sum_{n=0}^{\infty} (e^{p_i T} z^{-1})^n$$

$$= \sum_{i=1}^{N} A_i \left(\frac{1}{1 - e^{p_i T} z^{-1}} \right)$$

This is the required transfer function of digital filter.

Mapping of poles

(i) $\frac{1}{s - p_i} \longrightarrow \frac{1}{1 - e^{p_i T} z^{-1}}$

(ii) $\frac{s+a}{(s+a)^2 + b^2} \longrightarrow \frac{1 - e^{-aT} \cos(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}}$

(iii) $\frac{b}{(s+a)^2 + b^2} \longrightarrow \frac{e^{-aT} \sin(bT) z^{-1}}{1 - 2e^{-aT} \cos(bT) z^{-1} + e^{-2aT} z^{-2}}$

Q) Determine H(z) using IIM for the analog system function

$$H_a(s) = \frac{1}{(s+1)(s^2+s+2)}$$

Sol^n

Let, $H_a(s) = \frac{A}{(s+1)} + \frac{Bs+C}{s^2+s+2}$

or, $\frac{1}{(s+1)(s^2+s+2)} = \frac{A(s^2+s+2) + (Bs+2)(s+1)}{(s+1)(s^2+s+2)}$

or, $1 = (A+B)s^2 + (A+B+C)s + (2A+C)$

Comparing the coefficient, we get,

$A+B = 0$

$A+B+C = 0$

$2A+C = 1$

By solving, we get,

$A = 0.5$, $B = -0.5$, $C = 0$

Then, $H_a(s) = \frac{0.5}{s+1} - \frac{0.5s}{s^2+s+2}$

$= \frac{0.5}{s-(-1)} - \frac{0.5s}{(s+0.5)^2 + 1.75}$

$= \frac{0.5}{s-(-1)} - \frac{0.5s}{(s+0.5)^2 + (1.322)^2}$

$= \frac{0.5}{s-(-1)} - 0.5 \left[\frac{s+0.5-0.5}{(s+0.5)^2 + (1.322)^2} \right]$

$= \frac{0.5}{s-(-1)} - 0.5 \left[\frac{s+0.5}{(s+0.5)^2 + (1.322)^2} \right] + \left[\frac{0.5^2}{(s+0.5)^2 + (1.322)^2} \right]$

$= \frac{0.5}{s-(-1)} - 0.5 \left[\frac{s+0.5}{(s+0.5)^2 + (1.322)^2} \right] + \left[\frac{0.5^2 \times \frac{1.322}{1.322}}{(s+0.5)^2 + (1.322)^2} \right]$

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$$= \frac{-0.5}{s-(-1)} - 0.5\left[\frac{s+0.5}{(s+0.5)^2+(1.322)^2}\right] + 0.189\left[\frac{1.322}{(s+0.5)^2+(1.322)^2}\right]$$

$$= 0.5\left(\frac{1}{1-e^{-T}z^{-1}}\right) - 0.5\left[\frac{1-e^{-0.5T}\cos(1.32T)z^{-1}}{1-2e^{-0.5T}\cos(1.322T)z^{-1}+e^{-T}z^{-2}}\right]$$

$$+0.189\left[\frac{e^{-0.5T}\sin(1.322T)z^{-1}}{1-2e^{-0.5T}\cos(1.322T)z^{-1}+e^{-T}z^{-2}}\right]$$

(ii) BLT

Derivation:

Let us consider that the system function of Analog Filter is expressed as:

$$H_a(s) = \frac{b}{s+a} \quad \text{——— (i)}$$

The differential eqⁿ that describes the analog filter given by eqⁿ (i) is represented by

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$sY(s) + aY(s) = bX(s) \quad \text{——— (ii)}$$

Taking inverse laplace eqⁿ:

$$\frac{dy(t)}{dt} + ay(t) = bx(t) \quad \text{——— (iii)}$$

Now, integrating eqⁿ (iii) between $(nT-T)$ and nT

$$\int_{nT-T}^{nT} \frac{dy(t)}{dt} + \int_{nT-T}^{nT} ay(t) = \int_{nT-T}^{nT} bx(t) \quad \text{——— (iv)}$$

Using trapezoidal rule of integration in eqⁿ (iv),

$$y(nT) - y(nT-T) + \frac{aT}{2}y(nT) + \frac{aT}{2}y(nT-T) = \frac{bT}{2}x(nT) + \frac{bT}{2}x(nT-T)$$

Taking z-transform of above eqⁿ on both sides, we get,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{\frac{2}{T}\left[\frac{1-z^{-1}}{1+z^{-1}}\right]+a} \quad \text{——— (v)}$$

Now, comparing eqⁿ (i) and (v), we get,

$$s = \frac{2}{T}\left[\frac{1-z^{-1}}{1+z^{-1}}\right] \quad \text{——— (vi)} \quad \text{or, } s = \frac{2}{T}\left[\frac{z-1}{z+1}\right] \quad \text{——— (vii)}$$

The general characteristics of mapping $z = e^{sT}$ may be obtained by putting $s = \sigma + j\Omega$ and expressing the complex variable z in a polar form $z = re^{j\omega}$ in eqⁿ (vii), we get,

$$s = \frac{2}{T}\left[\frac{re^{j\omega}-1}{re^{j\omega}+1}\right]$$

$$= \frac{2}{T}\left[\frac{r^2-1}{1+r^2+2r\cos\omega} + j\frac{2r\sin\omega}{1+r^2+2r\cos\omega}\right] \quad \text{——— (viii)}$$

Let us consider for unity magnitude, $r=1$, then eqⁿ

Now, comparing eqⁿ (iii) with $s = \sigma + j\Omega$, we get,

$$\sigma = \frac{2}{T}\frac{r^2-1}{1+r^2+2r\cos\omega} \quad \text{——— (viii)}$$

$$\Omega = \frac{2r\sin\omega}{1+r^2+2r\cos\omega} \quad \text{——— (ix)}$$

Now, Let us consider for unity magnitude ($r=1$), then eqⁿ (iii) & (ix) becomes,

$$\Omega = \frac{2}{T}\left[\frac{2\sin\omega}{2(1+r\cos\omega)}\right] = \frac{2\tan\omega/2}{T}$$

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Q) The system function of analog filter is given as,

$$H_a(s) = \frac{s+0.1}{(s+0.1)^2+16}$$

Obtain the system function of digital filter using bit which is resonant at $\omega = \frac{\pi}{2}$.

Sol^n

Here, comparing with $H_a(s) = \frac{s+a}{(s+a)^2+\Omega^2}$,

$$\Omega = 4.$$

Using BLT,

$$\Omega = \frac{2}{T}\tan\frac{\omega}{2}$$

$$4 = \frac{2}{T}\tan\frac{\pi}{4}$$

$$T = \frac{2}{4} = 0.5 \text{ sec}$$

Now,

$$H(z) = H_a(s)\Big|_{s\to\frac{1}{T}\left[\frac{z-1}{z+1}\right]}$$

$$= \frac{s+0.1}{(s+0.1)^2+16}\Bigg|_{s\to 4\left[\frac{z-1}{z+1}\right]}$$

$$= \frac{4\left(\frac{z-1}{z+1}\right)+0.1}{\left[4\left(\frac{z-1}{z+1}\right)+0.1\right]^2+16}$$

(Simplify गर्नुपर्ने 1 marks काट्ने।)
Last मा गर्ने।

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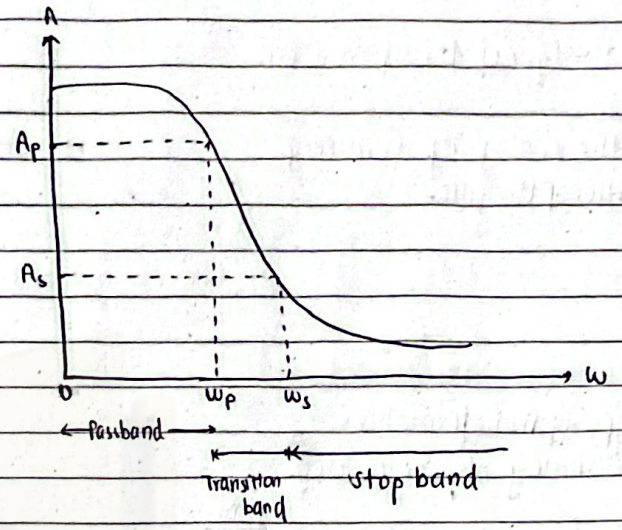
Prog Dig 3

Butterworth low pass filter



ω_p = passband frequency in rad/sec
ω_s = stop band frequency in rad/sec
K_p = Passband gain (dB) at frequency ω_p
K_s = stopband gain (dB) at frequency ω_s
$A_p = 10^{-K_p/20}$

$A_s = 10^{K_s/20}$

[Note: Always take K_p and K_s as negative. If in question K_p and K_s is negative, even then take K_p and K_s as negative]

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Imp²

Steps to design Butterworth Low Pass Filter.

Step 1: Choose the type of transformation.

step 2. ~~An~~ calculate the analog edge frequency.

3. ~~Decide the order of the filter.~~

BLT

Ω_p

1. Choose the type of transformation.
2. Calculate the analog edge frequency.

BLT	IIM
$\Omega_p = \frac{2}{T}\tan\frac{\omega_p}{2}$	$\Omega_p = \frac{\omega_p}{T}$
$\Omega_s = \frac{2}{T}\tan\frac{\omega_s}{2}$	$\Omega_s = \frac{\omega_s}{T}$

3. Decide the order of the filter.

$$N \geq \frac{1}{2}\frac{\log\left\{\left[\frac{1}{A_s^2}-1\right]/\left[\frac{1}{A_p^2}-1\right]\right\}}{\log\left[\frac{\Omega_s}{\Omega_p}\right]}$$

4. Calculate the analog cut-off frequency.

$$\Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2}-1\right]^{\frac{1}{2N}}}$$

5. When N is even

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k\Omega_c s + \Omega_c^2}$$

when N is odd,

$$H_a(s) = \frac{\Omega_c}{s+\Omega_c}\prod_{k=1}^{N-1/2} \frac{\Omega_c^2}{s^2 + b_k\Omega_c s + \Omega_c^2}$$

where, $b_k = 2\sin\left[\frac{(2k-1)\pi}{2N}\right]$

6. Using choosen transformation, the analog filter $H_a(s)$ is converted into $H(z)$.

Note: $A_p \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq \omega_p$

$|H[e^{j\omega}]| \leq A_s \quad \omega_s \leq \omega \leq \pi$

2559

Q) Using BLT, design a butterworth filter which satisfy the following conditions.

$0.8 \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$

$|H[e^{j\omega}]| \leq 0.2 \quad 0.6\pi \leq \omega \leq \pi$

Comparing with st. eqⁿ,

$A_p = 0.8 \quad \omega_p = 0.2\pi$

$A_s = 0.2 \quad \omega_s = 0.6\pi$

Let T = 1 sec.

PTO

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For BLT, $\Omega_p = \frac{2}{T}\tan\frac{\omega_p}{2}$ $\Omega_s = \frac{2}{T}\tan\frac{\omega_s}{2}$

$= 2\times\tan\frac{0.2\pi}{2}$ $= 2\times\tan\frac{0.6\pi}{2}$

$\therefore \Omega_p = 0.64$ $\therefore \Omega_s = 2.75$

Now,

$$N \geq \frac{1}{2}\frac{\log\left\{\left[\frac{1}{A_s^2}-1\right]/\left[\frac{1}{A_p^2}-1\right]\right\}}{\log[\Omega_s/\Omega_p]}$$

$$\geq \frac{1}{2}\frac{\log\left\{\left[\frac{1}{0.2^2}-1\right]/\left[\frac{1}{0.8^2}-1\right]\right\}}{\log[2.75/0.64]}$$

$N \geq 1.247$

$\therefore N \simeq 2$

Then, $\Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2}-1\right]^{1/2N}}$

$= \frac{0.64}{\left[\frac{1}{0.8^2}-1\right]^{1/4}}$

$= 0.739$

As N is even,

$b_1 = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2} = 1.414$

$$H_a(s) = \frac{0.739^2}{s^2 + 1.0455s + 0.739^2}$$

Transforming into H(z),

Now,

$$H(z) = H_a(s)\Big|_{s\to\frac{2}{T}\left(\frac{z-1}{z+1}\right)}$$

(Kp, Ks → +ve मानेको सबै Ap, As भन्दा सानो)

Butterworth — 1st page

2023spring

Q) Design a low pass butterworth filter using BLT satisfying the following constraints:

1. Passed band is frequency 0-400 Hz
2. stop band is frequency 2.1 - 4 KHz
3. Passed band ripple = 2 dB
4. stop band attenuation = 20 dB
5. sampling frequency = 10 KHz

Soln

f_p = 0-400 Hz = 400 Hz (सबैभन्दा high value)

f_s = 2.1-4 KHz = 2.1 KHz (सबैभन्दा low value)

$K_p = 2$ dB

$K_s = 20$ dB

$f_{sf} = 10$ KHz

$\therefore A_p = 10^{K_p/20} = 10^{-2/20} = 0.79$

$\therefore A_s = 10^{K_s/20} = 10^{-20/20} = 0.1$

$\omega_p = \frac{2\pi f_p}{f_{sf}} = \frac{2\pi\times 400}{10000} = \frac{4\pi}{50}$ rad/s

$\omega_s = \frac{2\pi f_s}{f_{sf}} = \frac{2\pi\times 2100}{10000} = \frac{21\pi}{50}$ rad/s.

freq. Hz मा दिना सबै, for calculating ω_p, ω_s, we need to normalize by dividing by sampling freq. (f_{sf}).

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Sampling freq. must be greater than two times max^m freq.
यदि दिएका छन् use, $f_{sf} \geq 2 f_{max}$ use गरर f_{sf} निकाल्ने,

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Now, we have, $T = \frac{1}{f_{sf}} = \frac{1}{10000}$ sec

For BLT,
Now, $\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$

$= 2 \times 10000 \times \tan\left(\frac{1}{2} \times \frac{4\pi}{50}\right)$

$= 2526.58$ rad/sec

$\Omega_s = \frac{2}{T} \tan \frac{\omega_s}{2}$

$= 2 \times 10000 \times \tan\left(\frac{1}{2} \times \frac{21\pi}{50}\right)$

$= 15513.59$ rad/sec

Then,

$$N \geq \frac{1}{2} \frac{\log\left[\left[\frac{1}{A_s^2} - 1\right] / \left[\frac{1}{A_p^2} - 1\right]\right]}{\log\left[\frac{\Omega_s}{\Omega_p}\right]}$$

$$= \frac{1}{2} \times \frac{\log\left(\left[\frac{1}{0.1^2} - 1\right] \Big/ \left[\frac{1}{0.79^2} - 1\right]\right)}{\log\left(\frac{15513.59}{2526.58}\right)}$$

$N \geq 1.4$

$\therefore N = 2$

Impulse invariance
bilinear transformation
elliptic filter
bessel filter
spectral transformation
} I

Date
Page

Now, $\Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2} - 1\right]^{1/2N}}$

$= \frac{2526.58}{\left[\frac{1}{0.79^2} - 1\right]^{\frac{1}{2\times 2}}}$

$\therefore \Omega_c = 2867.99$ rad/sec

When N is even,

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

so, $b_k = 2\sin\left[\frac{(2k-1)\pi}{2N}\right]$

$b_1 = 2\sin\left[\frac{2\times 1 - 1}{2\times 2}\pi\right]$ $\therefore b_1 = \sqrt{2}$

Then, $H_a(s) = \prod_{k=1}^{1} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$

$= \frac{2867.99^2}{s^2 + \sqrt{2} \times 2867.99 s + 2867.99^2}$

Transforming into H(z),

$H(z) = H_a(s)\Big|_{s \to \frac{2}{T}\left(\frac{z-1}{z+1}\right)}$

$= \frac{2867.99^2}{s^2 + 4055.99 s + 2867.99^2}\Big|_{s \to \frac{2}{T}\left(\frac{z-1}{z+1}\right)}$

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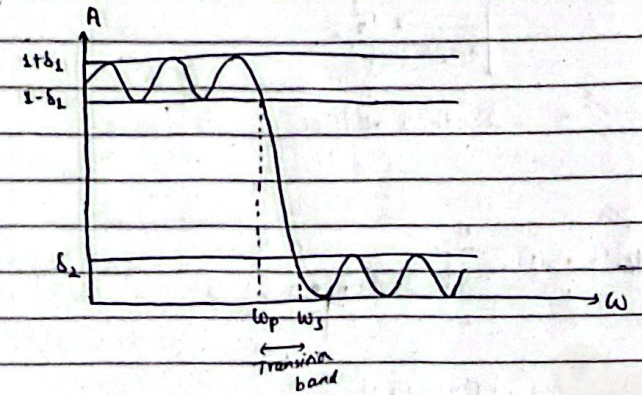
Unit-6

FIR Filter Design

1. Window Method



The magnitude Response is given by,

$$1-\delta_1 \le |H(e^{j\omega})| \le 1+\delta_1 \qquad 0 \le \omega \le \omega_p$$
$$|H(e^{j\omega})| \le \delta_2 \qquad \omega_s \le \omega \le \pi$$

Transition band $\Delta f = \frac{\omega_s - \omega_p}{2\pi}$

Length of filter $M = N = K\left(\frac{2\pi}{\omega_s - \omega_p}\right)$

$\omega_c = \frac{\omega_s + \omega_p}{2}$

width of main lobe $= K\left(\frac{2\pi}{M}\right)$

The impulse response of FIR Filter is given by,

$H_d(\omega) = e^{-j\omega\alpha}$ for $-\omega_c \le \omega \le \omega_c$,

$n = 0, 1, 2, \ldots M-1$

where, $\alpha = \frac{M-1}{2}$

$$h(n) = h_d(n) \times w(n)$$

Rectangular window

A rectangular window of length 'n' is represented by $w_R(n)$ and is expressed as,

$$w_R(n) = 1 \text{ for } n = 0, 1, \ldots M-1 \quad \text{——— (i)}$$
$$0 \text{ elsewhere}$$

Taking the fourier transform of eq^n (i), we get,

$$W_R(\omega) = \sum_{n=0}^{M-1} w_R(n) e^{-j\omega n}$$
$$= \sum_{n=0}^{M-1} 1 \cdot e^{-j\omega n}$$
$$= \sum_{n=0}^{M-1} (e^{-j\omega})^n \quad \text{——— (ii)}$$

$$\left(\sum_{k=N_1}^{N_2} a^k = \frac{a^{N_1} - a^{N_2+1}}{1-a}\right)$$

Now, expanding the finite series of eq^n (ii), we get,

$$W_R(\omega) = \frac{(e^{-j\omega})^0 - (e^{-j\omega})^{M-1+1}}{1-e^{-j\omega}}$$
$$= \frac{1-(e^{-j\omega})^M}{1-e^{-j\omega}}$$
$$= \frac{e^{\frac{-j\omega M}{2}} \cdot e^{\frac{j\omega M}{2}} - e^{\frac{-j\omega M}{2}} \cdot e^{\frac{-j\omega M}{2}}}{e^{\frac{-j\omega}{2}} \cdot e^{\frac{j\omega}{2}} - e^{\frac{-j\omega}{2}} \cdot e^{\frac{-j\omega}{2}}}$$
$$= \frac{e^{\frac{-j\omega M}{2}} \left[e^{\frac{j\omega N}{2}} - e^{\frac{-j\omega N}{2}}\right]}{e^{\frac{-j\omega}{2}} \left[e^{\frac{j\omega}{2}} - e^{\frac{-j\omega}{2}}\right]}$$

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$\sin\omega = \frac{e^{j\omega} - e^{-j\omega}}{2j}$

$$= \frac{e^{-j\omega\frac{M}{2}}\left[2j\sin\frac{\omega M}{2}\right]}{e^{-j\frac{\omega}{2}}\left[2j\sin\frac{\omega}{2}\right]}$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)}\left[\frac{\sin\frac{\omega M}{2}}{\sin\frac{\omega}{2}}\right]$$

The magnitude response is given by,

$$|W_R(\omega)| = \frac{\left|\sin\frac{\omega M}{2}\right|}{\left|\sin\frac{\omega}{2}\right|}$$

1) Hamming window

$$w_H(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right) \quad \text{for } n = 0, 1, 2 \ldots M-1$$

$$0 \quad \text{elsewhere}$$

2) Hanning window

$$w_{Hn}(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right) \quad \text{for } n = 0, 1, \ldots M-1$$

$$0 \quad \text{elsewhere}$$

Attenuation and transition width:

Name of window	width of main lobe (Transition width)	Stopband Attenuation
1) Rectangular	$\frac{4\pi}{M+1}$	-21 dB
~~2) Hamm~~		
2) Hanning	$\frac{8\pi}{M}$	-44 dB
3) Hamming	$\frac{8\pi}{M}$	-53 dB

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2024 fall

Q) Design a FIR filter with (Using Hanning window and M=9)

$$H_d(e^{j\omega}) = 1 \quad -\pi/6 \le \omega \le \pi/6$$

$$0 \quad \text{otherwise}$$

Sol^n

We have, $h(n) = h_d(n) \times w(n)$

$$\omega_c = \frac{\pi}{6}$$

Here, $h_d(n) = \frac{1}{2\pi}\int_{\omega_c}^{\omega_c} H_d(\omega)\, e^{j\omega n}\, d\omega$

$$= \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\left[\frac{e^{j\omega n}}{jn}\right]_{-\pi/6}^{\pi/6}$$

$$= \frac{1}{\pi n}\left[\frac{e^{j\frac{n\pi}{6}} - e^{-jn\pi/6}}{2j}\right]$$

$\sin n = \frac{e^{jn} - e^{-jn}}{2j}$

$$h_d(n) = \frac{1}{\pi n}\sin\left(\frac{n\pi}{6}\right) \quad (n \neq 0) \rightarrow \text{zero by } \tfrac{0}{0} \text{ form}$$

For $n = 0$, $h_d(n) = \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} e^{0}\, d\omega$

$$= \frac{1}{2\pi}\left(\frac{\pi}{6} + \frac{\pi}{6}\right)$$

$$= \frac{1}{2\pi} \times \frac{2\pi}{6} = \frac{1}{6}$$

For hanning window, $w(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{m-1}\right)$ for $n = 0, 1, 2, \ldots 8$

$$h(0) = h_d(0) \times w(0)$$

$$= \frac{1}{6} \times 0 = 0$$

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$h(n) = h(M-1-n)$

Hence, $h(n) = h_d(n) \times w_{Hm}(n)$

$$= \begin{bmatrix} \frac{1}{\pi n}\sin\left(\frac{n\pi}{6}\right) & \text{for } n \neq 0 \\ 1/6 & \text{for } n = 0 \end{bmatrix} \begin{bmatrix} 0.5 - 0.5\cos\left(\frac{2\pi n}{8}\right) \\ 0.5 - 0.5\cos\left(\frac{2\pi n}{8}\right) \end{bmatrix}$$

n	$h_d(n) = \frac{\sin(n\pi/6)}{n\pi}$	$w(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{8}\right)$	$h(n)$
0	0.1667	0	0
1	0.1592	0.1465	
2	0.1378	0.5	
3	0.1061	0.8535	
4	0.0689	1	
5			
6			
7			
8			

symmetric

Properties of Filters

~~# Properties~~

1. Chebyshev Filter

- Chebyshev Filter has a non-flat magnitude response.
- It exhibits ripples in the frequency response.
- It provides a faster roll-off than Butterworth filter.
- The transition band is narrow.
- Phase response is non-linear.
- Requires lower order than Butterworth for same specifications.
- Two types exist: (i) Type-I (passband ripple)
  (ii) Type-II (stopband ripple)
- Amplitude distortion occurs due to ripple.

2. Elliptic (Cauer) Filter

- Elliptic filter has ripples in both passband and stopband.
- It provides the sharpest cutoff among all classical filters.
- The transition band is minimum.
- Requires the smallest filter order.
- Contains zeros in both passband and stopband.
- Phase response is highly non-linear.
- Design complexity is high.
- Frequency selectivity is very high.

3. Bessel Filter

- Bessel Filter provides linear phase response.
- Group delay is constant in passband.
- There are no ripples in passband or stopband.
- Frequency roll-off is very slow.
- Poor frequency selectivity.
- Preserves waveform shape.
- Exhibits minimum overshoot and ringing.
- Time-domain performance is best.

4. Spectral Transformation

- Spectral transformation converts a low-pass prototype filter into other filters.
- Simplifies the filter design process.
- Used in analog and digital IIR filter design.
- Helps avoid redesigning filters from scratch.
- Preserves the filter order.

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Linear phase → FIR filter

- Used to obtain HPF, BPF and BSF from LPF.
- Achieved using frequency substitution.
- common in classical filter theory.

25 yrs Q) Design a FIR system to meet the following specifications.
Passed band is frequency = 2 KHz
Stop band is frequency = 5KHz
Stop band Attenuation = 42dB
sampling Frequency = 20KHz

Sol^n

As the stop band attenuation is 42 dB, Hence Hanning window must be considered.

$f_p = 2KHz$

$f_s = 5KHz$

$f_{sf} = 20KHz$

$\omega_p = \frac{2\pi f_p}{f_{sf}} = \frac{2\pi \times 2}{20} = \frac{\pi}{5}$

$\omega_s = \frac{2\pi f_s}{f_{sf}} = \frac{2\pi \times 5}{20} = \frac{\pi}{2}$

Now, $\omega_c = \frac{\omega_s + \omega_p}{2} = \frac{\pi/2 + \pi/5}{2} = \frac{7\pi}{20}$

We have, width of main lobe $= K\left(\frac{2\pi}{M}\right)$

or, $\frac{8\pi}{M} = K\left(\frac{2\pi}{M}\right)$

$\therefore K = 4$

Design FIR : ω_c M window
calculate

Length of filter, $M = K\left(\frac{2\pi}{\omega_s - \omega_p}\right)$

$= 4 \times \left(\frac{2\pi}{\pi/2 - \pi/5}\right)$

$= \frac{80}{3}$

$= 26.67 \approx 27$ {integer पूर्णांक}

$\alpha = \frac{M-1}{2} = \frac{27-1}{2} = 13$

We have, $h(n) = h_d(n) \times w_{HN}(n)$

$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega) e^{j\omega n} d\omega$ {$H_d(\omega)$ दिएको छैन use $e^{-j\omega\alpha}$}

$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega$

$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega$

$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-13)} d\omega$

$= \frac{1}{2\pi}\left[\frac{e^{j\omega(n-13)}}{j(n-13)}\right]_{-\omega_c}^{\omega_c}$

$= \frac{1}{\pi(n-13)}\left[\frac{e^{j\omega_c(n-13)} - e^{-j\omega_c(n-13)}}{2j}\right]$

$= \frac{1}{\pi(n-13)} \sin\omega_c(n-13) \quad (n \neq 13)$

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For n=13,

$$h_d(13) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} d\omega = \frac{1}{2\pi} \times 2\omega_c$$

$$= \frac{7\pi}{20\pi}$$

$$= \frac{7}{20}$$

Hence,

$$h(n) = h_d(n) \times w_{Hm}(n)$$

$$w_{Hm}(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right)$$

$$= \begin{bmatrix} \frac{1}{\pi(n-13)} \sin\frac{7\pi}{20}(n-13) & \text{for } n \neq 13 \\ 7/20 & \text{for } n = 13 \end{bmatrix} \begin{bmatrix} 0.5 - 0.5\cos\frac{2\pi n}{26} \\ 0.5 - 0.5\cos\frac{2\pi n}{26} \end{bmatrix}$$

Ans

table बनाउने तरिका अर्को,

Q) Design a low pass FIR with 7 coefficients for the following specifications:

Passband freq = 300 Hz

Sampling freq = 1kHz

Use Hanning window.

Sol^n

$f_p = 300$ Hz $M = 7$ (7 coefficients)

$f_{sf} = 1000$ Hz

$$\omega_p = \frac{2\pi f_p}{f_{sf}} = \frac{2\pi \times 300}{1000} = \frac{3\pi}{5}$$

Now, $\omega_s = \omega_p + T.\omega.$

$$= \frac{3\pi}{5} + \frac{8\pi}{M}$$

$$= \frac{3\pi}{5} + \frac{8\pi}{7}$$

$$\omega_s = \frac{61\pi}{35}$$

Flowchart of Reverz algorithm

$$\omega_c = \frac{\omega_s + \omega_p}{2}$$

$$= \frac{61\pi/35 + 3\pi/5}{2}$$

$$\therefore \omega_c = \frac{41\pi}{35}$$

$$\alpha = \frac{M-1}{2} = \frac{7-1}{2} = 3$$

Now, $h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega) e^{j\omega n} d\omega$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n - 3j\omega} d\omega$$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-3)} d\omega$$

$$= \frac{1}{2\pi}\left[\frac{e^{j\omega(n-3)}}{j(n-3)}\right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{\pi(n-3)}\left[\frac{e^{j\omega_c(n-3)} - e^{-j\omega_c(n-3)}}{2j(n-3)}\right]$$

$$= \frac{1}{\pi(n-3)} \sin\omega_c(n-3)$$

$$= \frac{1}{\pi(n-3)} \sin\frac{41\pi}{35}(n-3) \quad \text{for } n \neq 3$$

For n=3, $h_d(3) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(3-3)} d\omega$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} d\omega = \frac{1}{2\pi} \times 2 \times \frac{41\pi}{35} = \frac{41}{35}$$

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Hence,

$$H(n) = H_d(n) \times W_{HN}(n)$$

$$= \begin{cases} \frac{1}{\pi(n-3)} \sin\frac{41\pi}{35}(n-3) & \text{for } n \neq 3 \\ 41/35 & \text{for } n = 3 \end{cases} \times \left[0.05 - 0.5\cos\frac{2\pi n}{6}\right]$$

Kaiser Window

In Kaiser window,

$$w(n) = \left[\frac{I_0\left\{\beta\left[1-\left(\frac{n-\alpha}{\alpha}\right)^2\right]^{1/2}\right\}}{I_0(\beta)}\right] \quad \text{for } n = 0, 1 \cdots M$$

Here, $\alpha = \frac{M}{2}$

And,

$$I_0(n) = 1 + \frac{0.25x^2}{(1!)^2} + \frac{(0.25x^2)^2}{(2!)^2} + \frac{(0.25x^2)^3}{(3!)^2} + \frac{(0.25x^2)^4}{(4!)^2} + \cdots$$

Imp 2 pg.

Design Steps

1. Choose the optimum value of δ.

$$\delta = \min(\delta_p, \delta_s)$$

$$\delta_p = 10^{-0.05A_s}$$

$$\delta_s = \frac{10^{0.05A_p} - 1}{10^{0.05A_p} + 1}$$

$$A_p = 20\log_{10}\left(\frac{1+\delta_p}{1-\delta_p}\right)$$

$$A_s = -20\log_{10}(\delta_s)$$

2. Calculate Attenuation A.

$$A = -20\log_{10}(\delta) \text{ dB}$$

3. Calculate parameter β.

$$\beta = \begin{cases} 0.1102(A-8.7) & \text{for } A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21) & \text{for } 21 \le A \le 50 \\ 0 & \text{for } A < 21 \end{cases}$$

4. The length of the filter (for kaiser window) is M+1. And, M is calculated by

$$M = \frac{A-8}{2.285\,\Delta w}$$

where, $\Delta w = w_s - w_p$

5. Determine w_c $\quad \therefore w_c = \frac{w_s + w_p}{2}$

6. Now, select the desired impulse response.

Then, $H(n) = h_d(n) \times w(n)$

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Numerical गरेको फरक ?

Q) Design a linear FIR filter to meet the following specifications.

$0.99 \le |H(e^{j\omega})| \le 1.01$ for $0 \le \omega \le 0.19\pi$

$|H(e^{j\omega})| \le 0.01$ for $0.21\pi \le \omega \le \pi$

Soln

$1 - \delta_1 = 0.99$

$\therefore \delta_1 = 0.01$

Note: $\delta_1 = \delta_p$, $\delta_2 = \delta_s$

$\delta_2 = 0.01$

$\omega_p = 0.19\pi$

$\omega_s = 0.21\pi$

Now, $\delta = \min(\delta_p, \delta_s)$

$= \min(\delta_1, \delta_2)$

$= 0.01$

Then, $A = -20\log_{10}(\delta)$

$= -20\log_{10}(0.01)$

$= 40\,dB$

Now, For $A = 40$,

$\beta = 0.5842(A-21)^{0.4} + 0.07886(A-21)$

$= 0.5842(40-21)^{0.4} + 0.07886(40-21)$

$= 3.395$

Then, $M = \frac{A-8}{2.285\,\Delta\omega}$

$= \frac{40-8}{2.285 \times (0.21\pi - 0.19\pi)}$

$\therefore M = 222.886$

⚠ M should be integer

Then, $\omega_c = \frac{\omega_c + \omega_p}{2}$

$= \frac{0.21\pi + 0.19\pi}{2}$

$\therefore \omega_c = \pi/5$

Here, $h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega) e^{j\omega n} d\omega$

$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega$

$= \frac{1}{2\pi}\left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\right]_{-\omega_c}^{\omega_c}$

$= \frac{1}{\pi(n-\alpha)} \sin\omega_c(n-\alpha)^{-\omega_c} \quad n \neq \alpha$

Here, $\alpha = \frac{M-1}{2}$

$= \frac{1}{\pi\left[n - \left(\frac{M-1}{2}\right)\right]} \times \sin\omega_c\left[n - \left(\frac{M-1}{2}\right)\right] \quad n \neq \frac{M-1}{2}$

For kaiser window, the length of filter = M+1,

$h_d(n) = \frac{1}{\pi\left[n - \frac{M+1-1}{2}\right]} \sin\omega_c\left[n - \frac{M+1-1}{2}\right] \quad n \neq \frac{M+1-1}{2}$

$= \frac{1}{\pi\left[n - \frac{M}{2}\right]} \sin\omega_c\left[n - \frac{M}{2}\right] \quad n \neq \frac{M}{2}$

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$$= \frac{1}{\pi(n-111.5)} \sin[0.2\pi(n-111.5)] \quad \text{for } n \neq 111.5$$

For 111.5,

$h_d(n) = 0.2$

Now,

$$h(n) = h_d(n) \times w(n)$$

$$= \left[\frac{1}{\pi(n-111.5)} \sin[0.2\pi(n-111.5)]\right] \left[\frac{I_0\left\{3.395\left[1-\left(\frac{n-111.5}{111.5}\right)^2\right]^{1/2}\right\}}{I_0(3.395)}\right] \quad \text{for } n \neq 111.5$$

$$\left[0.2\right] \left[\frac{I_0\left\{3.395\left[1-\left(\frac{n-111.5}{111.5}\right)^2\right]^{1/2}\right\}}{I_0(3.395)}\right] \quad \text{for } n = 111.5$$

Where,

$$I_0(x) = 1 + \frac{0.25x^2}{(1!)^2} + \frac{(0.25x^2)^2}{(2!)^2} + \frac{(0.25x^2)^3}{(3!)^2} + \frac{(0.25x^2)^4}{(4!)^2} + \ldots$$

Frequency sampling method

Derivation/steps:

Let us consider that FIR filter has to be designed whose desired freq. response is denoted by,

$$H_d(\omega) = e^{-j\omega\alpha} \quad 0 \le \omega \le \omega_c$$

$$0 \quad \text{otherwise} \quad \text{——— (i)}$$

where, $\alpha = \frac{M-1}{2}$

This freq. response is sampled uniformly at M points, such frequency samples are given as,

$$\omega = \frac{2\pi k}{M} \quad \text{when } k = 0, 1, \ldots M-1 \quad \text{——— (ii)}$$

Such sample desired frequency response is DFT which is denoted by $H(k)$ such that

$$H(k) = H_d(\omega)\Big|_{\omega \to \frac{2\pi k}{M}} \quad \text{——— (iii)}$$

By taking inverse DFT of $H(k)$, $h(n)$ is obtained which is given by:

$$h(n) = \frac{1}{M} \sum_{k=0}^{M-1} H(k) e^{\frac{j2\pi kn}{M}} \qquad n = 0, 1, 2 \ldots M-1$$

The unit sample response is

$$h(n) = \frac{1}{M}\left\{H(0) + 2\sum_{k=1}^{\frac{M-1}{2}} \mathrm{Re}\left[H(k) e^{\frac{j2\pi kn}{M}}\right]\right\} \quad \text{if M is odd}$$

and $$h(n) = \frac{1}{M}\left\{H(0) + 2\sum_{k=1}^{\frac{M}{2}-1} \mathrm{Re}\left[H(k) e^{\frac{j2\pi kn}{M}}\right]\right\} \quad \text{if M is even}$$

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Hence, unit sample response of FIR filter of length M is obtained using frequency sampling technique.

Q) Design a low pass FIR filter using freq. sampling technique having cut-off frequency of $\frac{\pi}{2}$ rad/s.

The filter should have linear phase and length of filter is 17.

Soln

$$\omega_c = \frac{\pi}{2}$$

$$H_d(\omega) = e^{-j\omega\alpha} \qquad 0 \le \omega \le \omega_c \quad \text{—— (i)}$$

where,

$$\alpha = \frac{M-1}{2} = \frac{17-1}{2} = 8$$

Then, eqn (i) becomes,

$$H_d(\omega) = e^{-j8\omega} \qquad 0 \le \omega \le \frac{\pi}{2} \quad \text{—— (ii)}$$

The sampled version of eqn (ii) can be given by,

$$H(k) = H_d(\omega)\Big|_{\omega \to \frac{2\pi k}{17}} \qquad 0 \le \omega \le \frac{\pi}{2}$$

$$= e^{-j8\omega}\Big|_{\omega \to \frac{2\pi k}{17}} \qquad 0 \le \omega \le \frac{\pi}{2}$$

$$= e^{-j8\times\frac{2\pi k}{17}} \qquad 0 \le \frac{2\pi k}{17} < \frac{\pi}{2}$$

$$= e^{-j\frac{16\pi k}{17}} \qquad 0 \times \frac{17}{2\pi} \le \frac{2\pi k}{17} \times \frac{17}{2\pi} < \frac{\pi}{2} \times \frac{17}{2\pi}$$

$$= e^{-j\frac{16\pi k}{17}} \qquad 0 \le k \le 5$$

Then, $h(n) = \frac{1}{M}\sum_{k=1}^{M-1} H(k)\, e^{\frac{j2\pi kn}{M}}$ for $n = 0, 1 \ldots 16$

Here, M = 17 (odd)

$$\therefore h(n) = \frac{1}{17}\left\{H(0) + 2\sum_{k=1}^{8} \mathrm{Re}\left[H(k)\, e^{\frac{j2\pi kn}{17}}\right]\right\} \text{ for } n = 0, 1, \ldots 16$$

$$= \frac{1}{17}\left\{1 + 2\sum_{k=1}^{5} \mathrm{Re}\left[e^{-j\frac{16\pi k}{17}} \cdot e^{\frac{j2\pi kn}{17}}\right]\right\} \text{ for } n = 0, 1 \ldots 16$$

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$$= \frac{1}{17}\left\{1 + 2\sum_{k=1}^{5} \mathrm{Re}\left[e^{\frac{j2\pi k}{17}(n-16)}\right]\right\} \text{ for } n = 0, 1 \ldots 16$$

$$\therefore h(n) = \frac{1}{17}\left\{1 + 2\sum_{k=1}^{5} \cos\left(\frac{2\pi k}{17}(n-16)\right)\right\} \text{ for } n = 0, 1, 16$$

$$\left[\because e^{j\eta} = \cos\eta + j\sin\eta\right]$$

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